

BFS Traversal

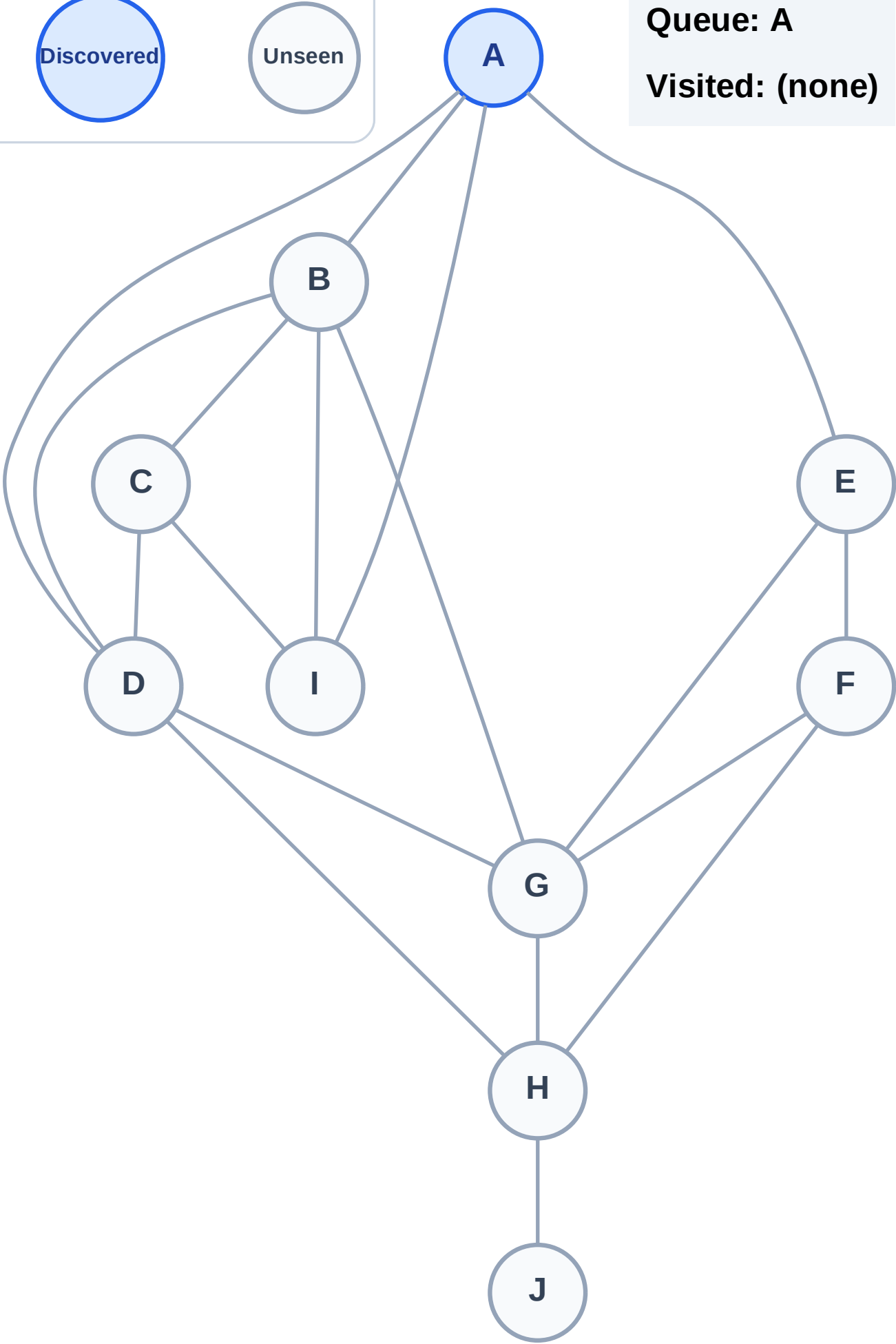
Step 1: Start at A

Legend



Queue: A

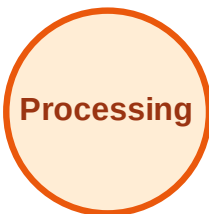
Visited: (none)



BFS Traversal

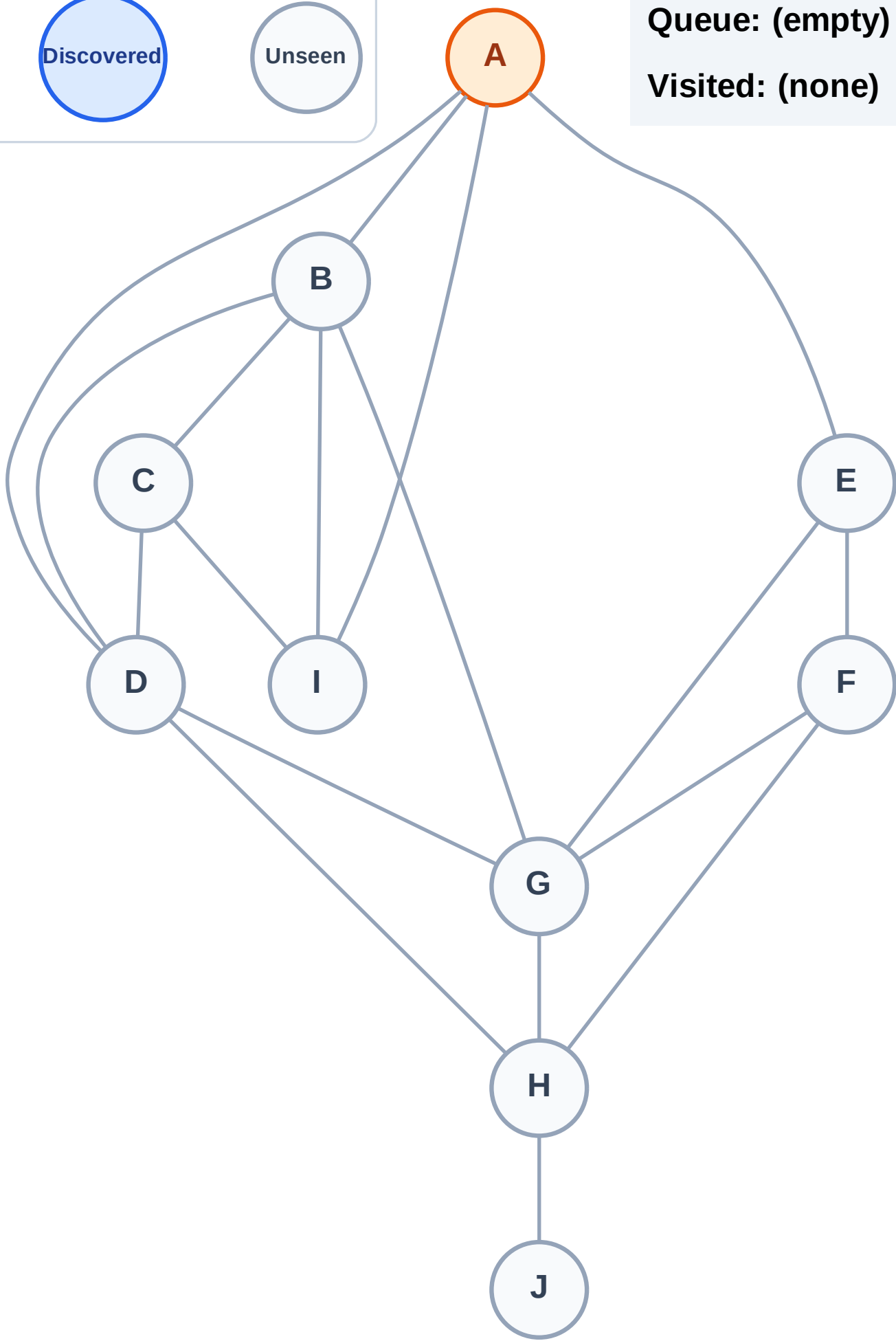
Step 2: Process node A

Legend



Queue: (empty)

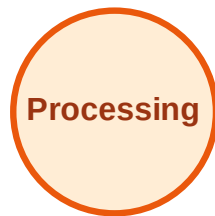
Visited: (none)



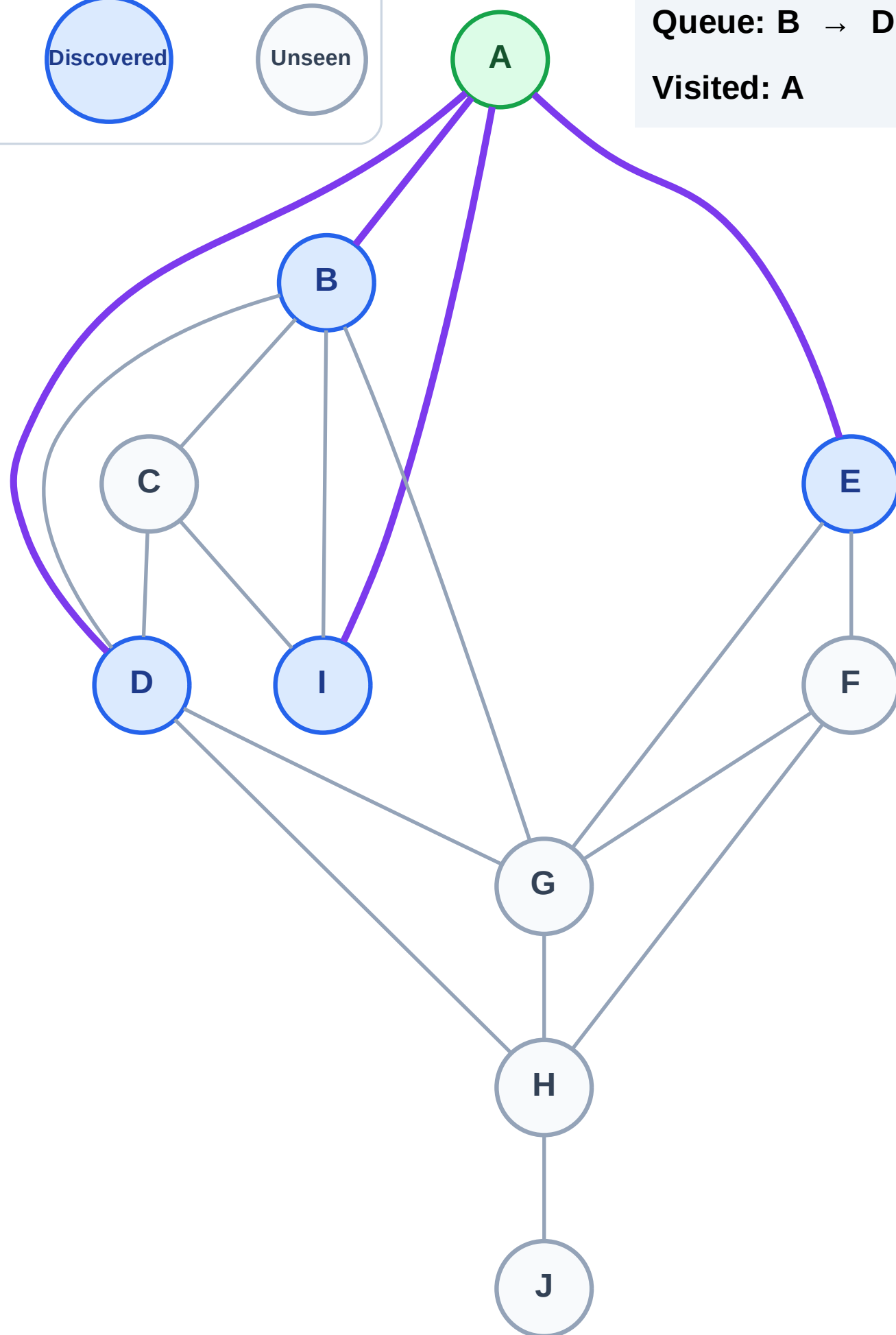
BFS Traversal

Step 3: Discover B, D, E, I from A

Legend



Queue: B → D → E → I
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

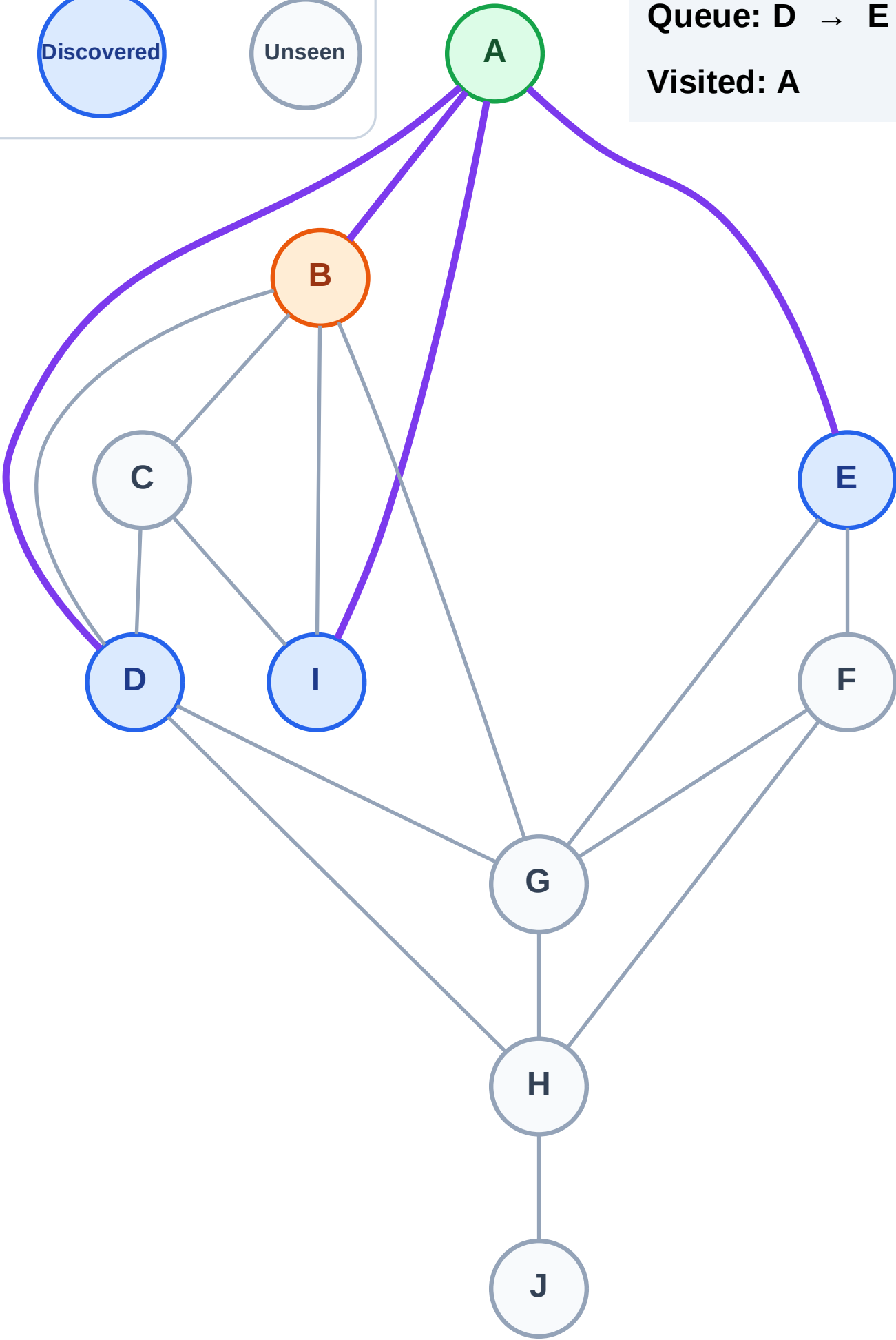
Processing

Discovered

Unseen

Queue: D → E → I

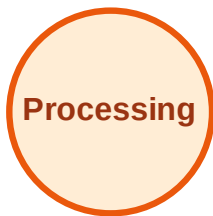
Visited: A



BFS Traversal

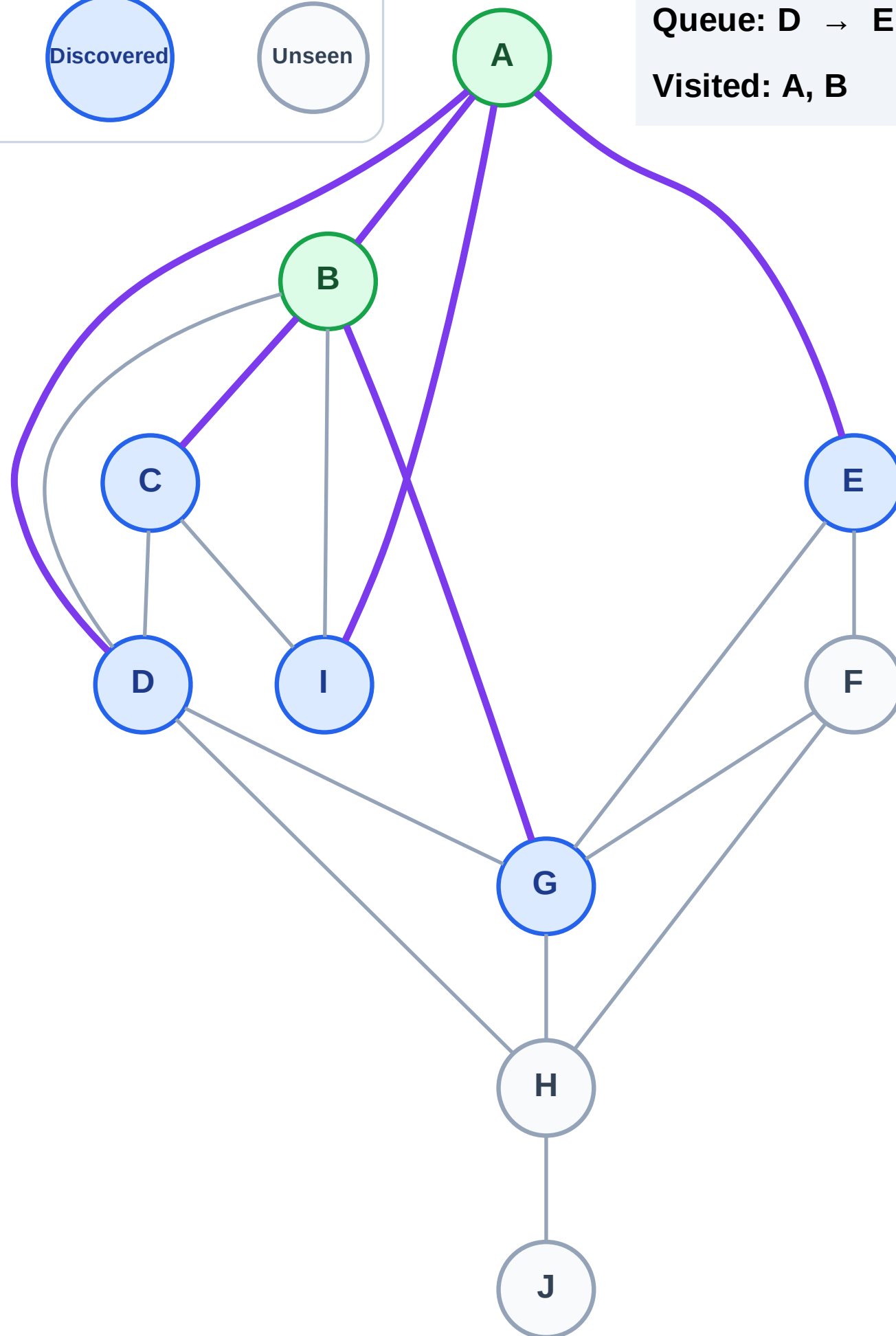
Step 5: Discover C, G from B

Legend



Queue: D → E → I → C → G

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

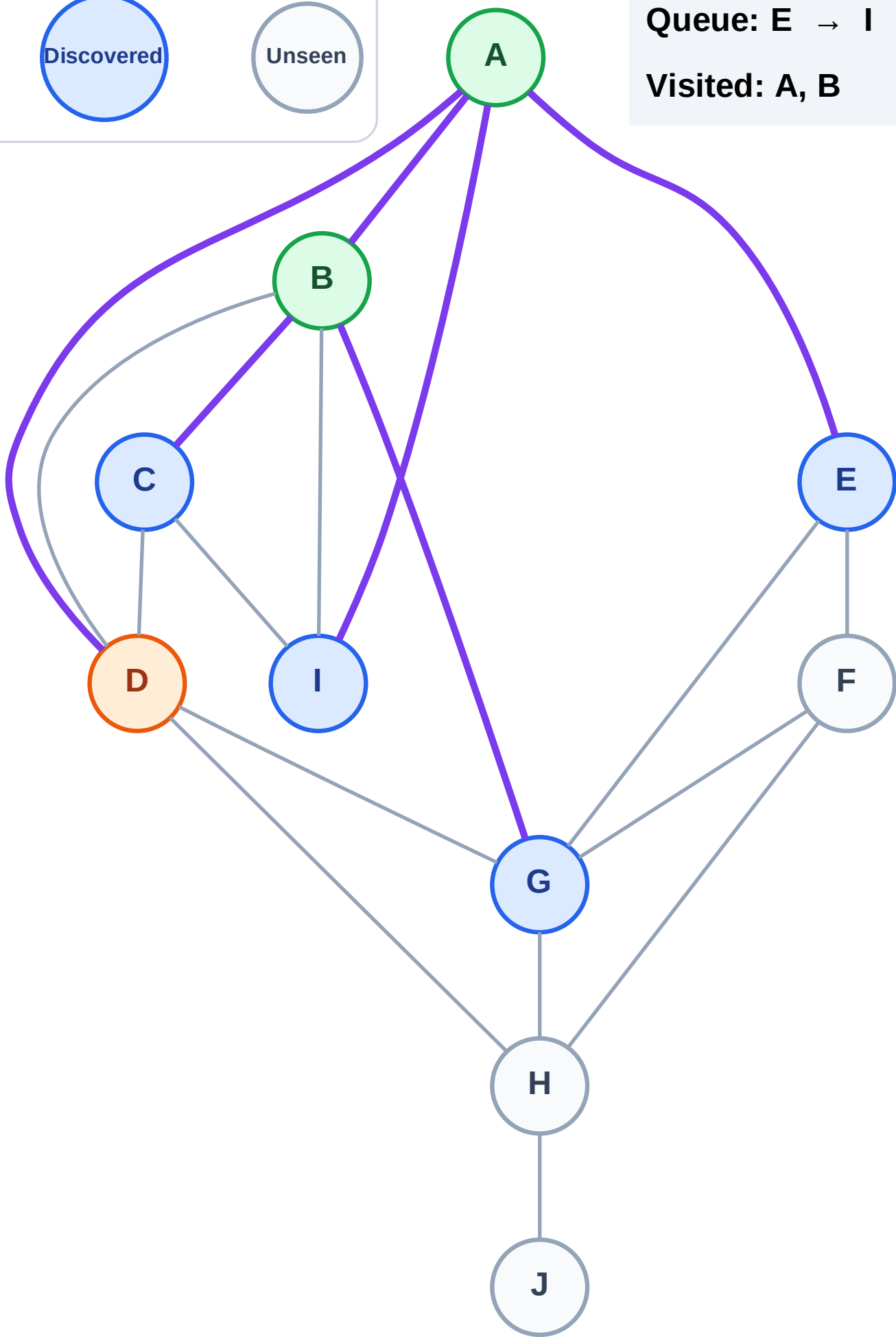
Processing

Discovered

Unseen

Queue: E → I → C → G

Visited: A, B



BFS Traversal

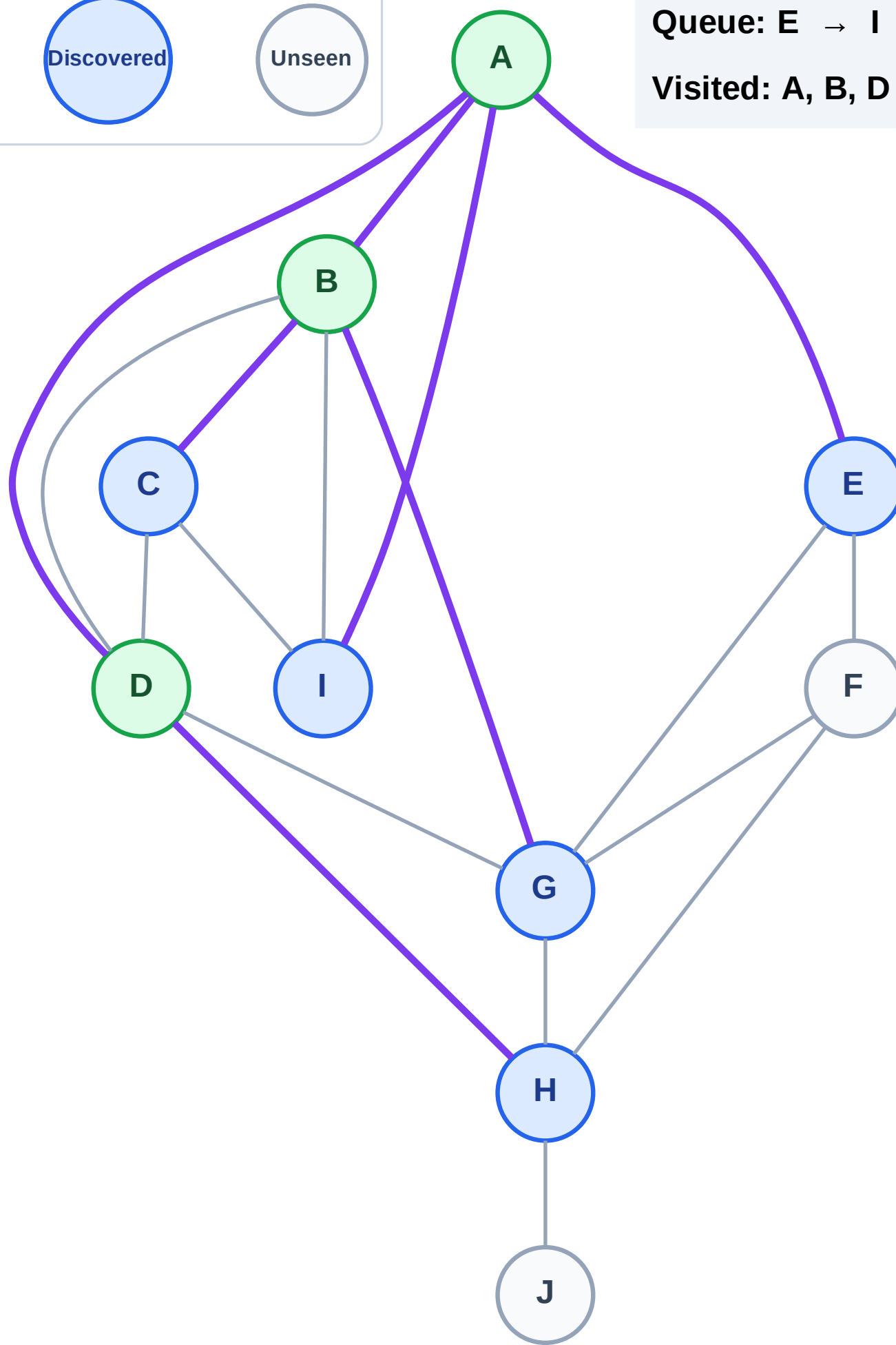
Step 7: Discover H from D

Legend



Queue: E → I → C → G → H

Visited: A, B, D



BFS Traversal

Step 8: Process node E

Legend

Complete

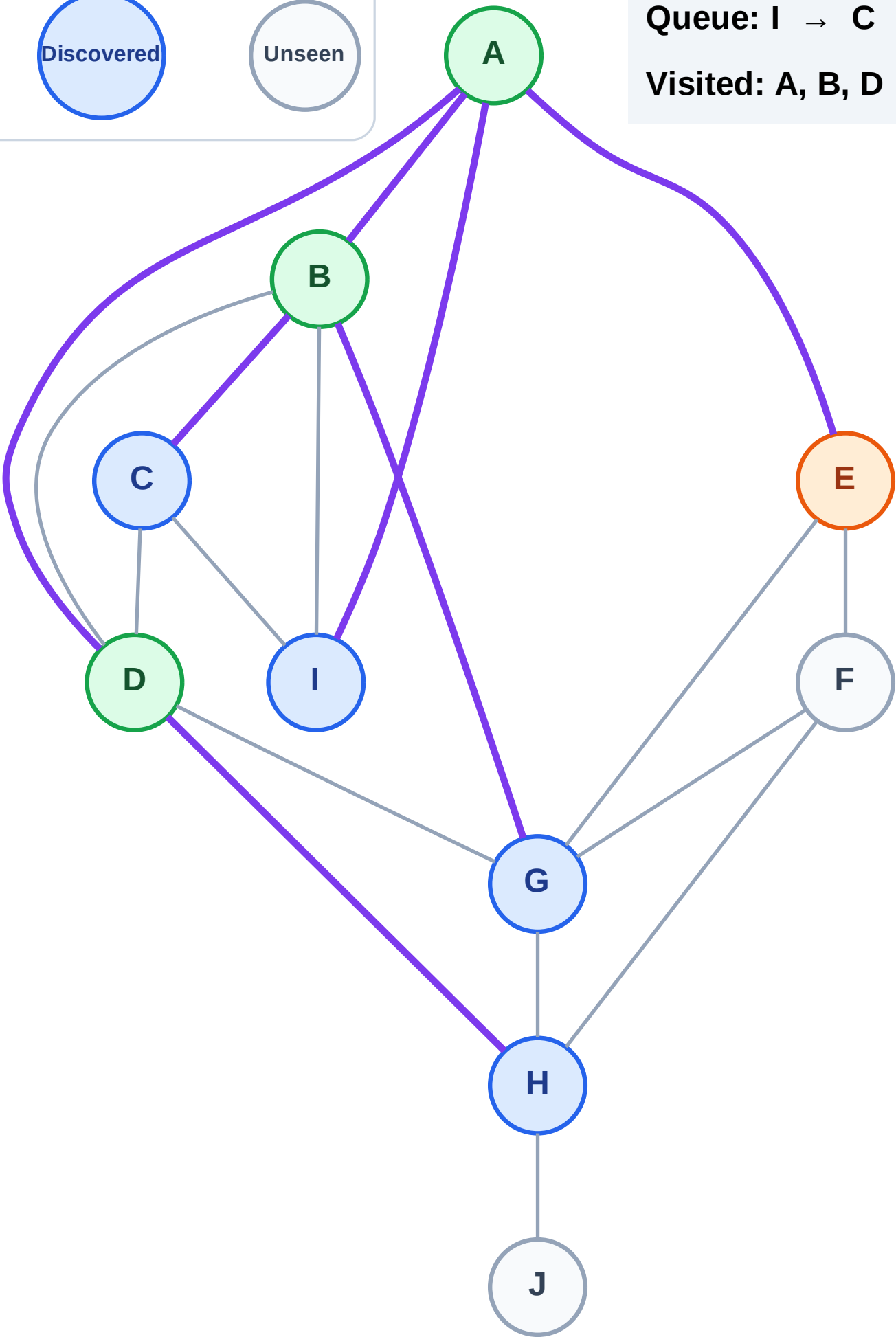
Processing

Discovered

Unseen

Queue: I → C → G → H

Visited: A, B, D



BFS Traversal

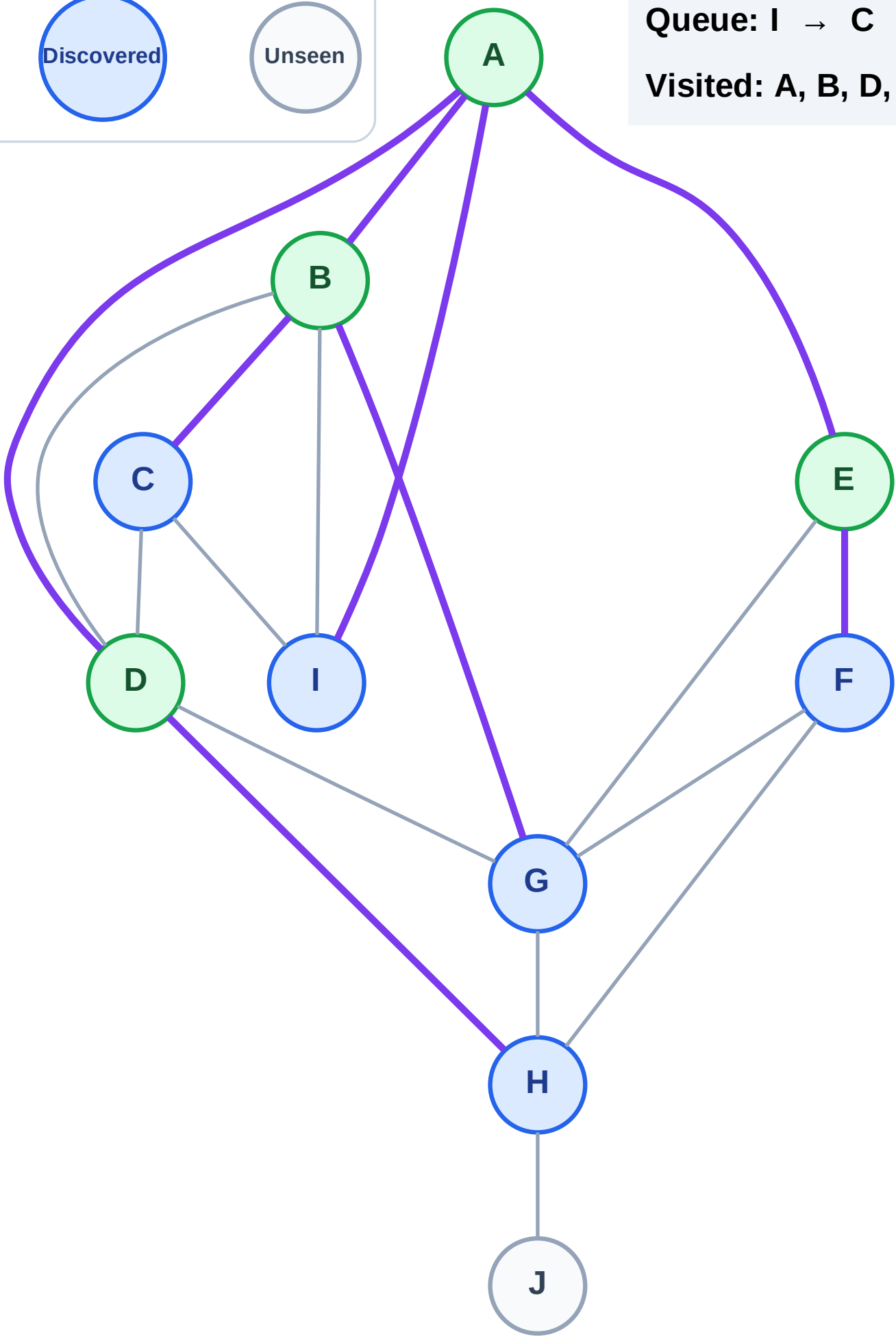
Step 9: Discover F from E

Legend



Queue: I → C → G → H → F

Visited: A, B, D, E



BFS Traversal

Step 10: Process node I

Legend

Complete

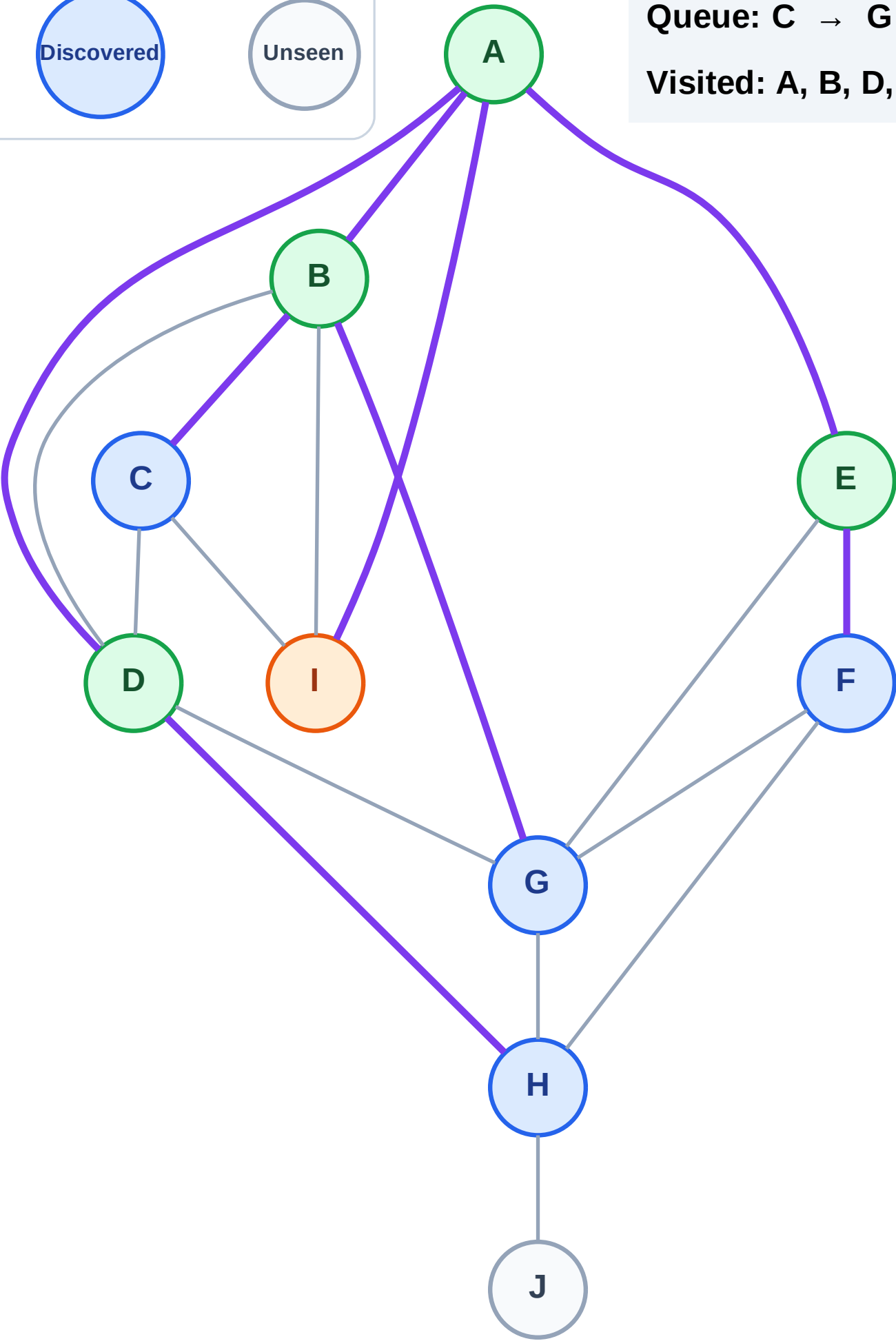
Processing

Discovered

Unseen

Queue: C → G → H → F

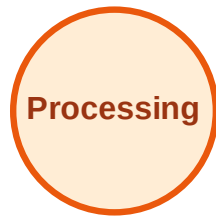
Visited: A, B, D, E



BFS Traversal

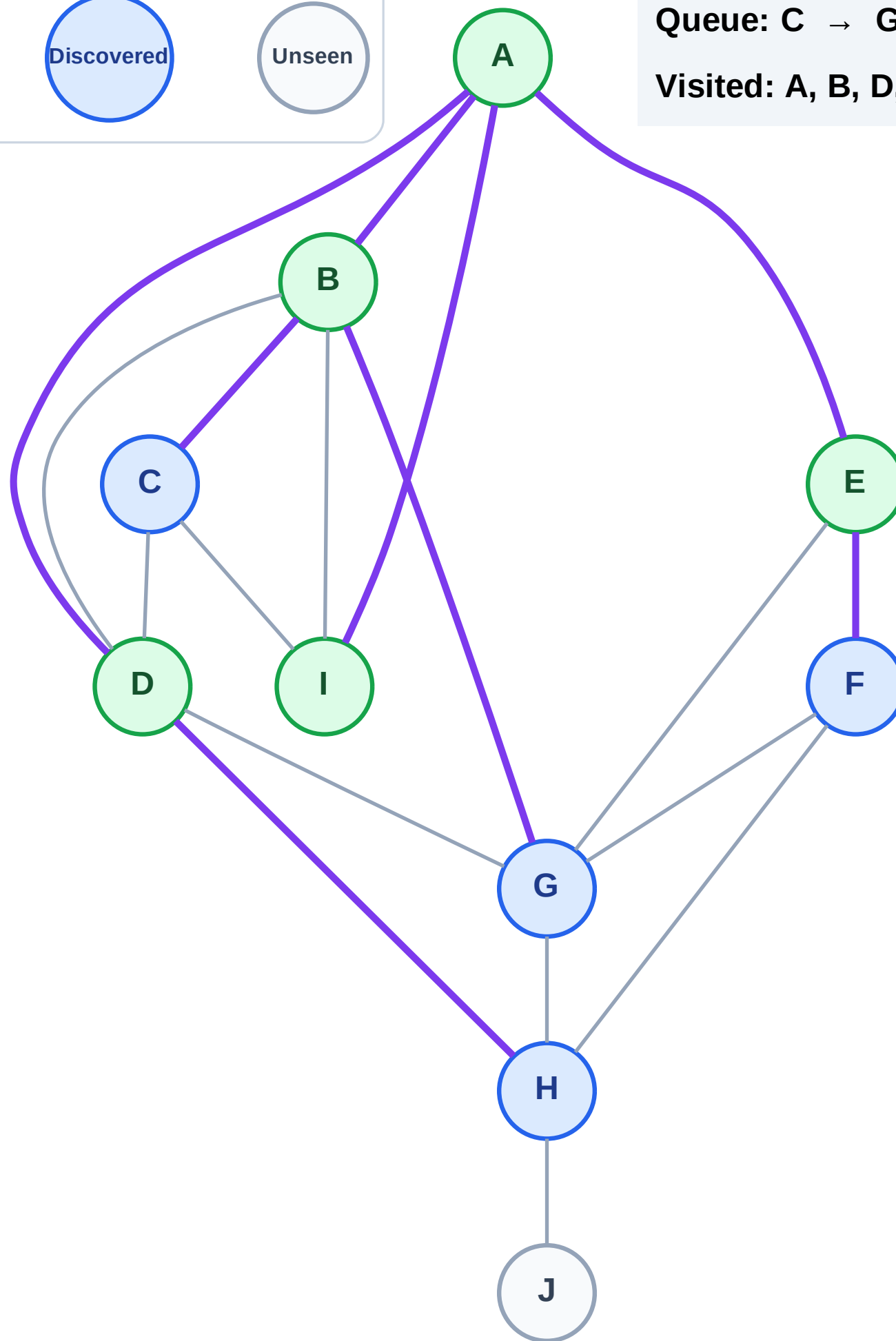
Step 11: No new neighbors from I

Legend



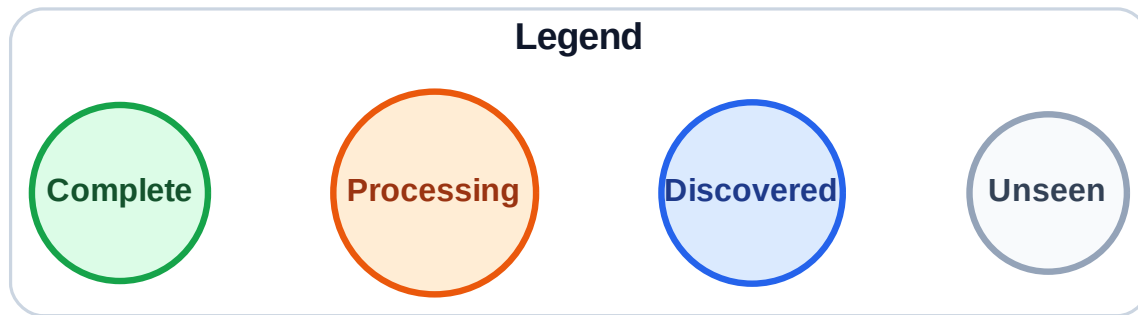
Queue: C → G → H → F

Visited: A, B, D, E, I

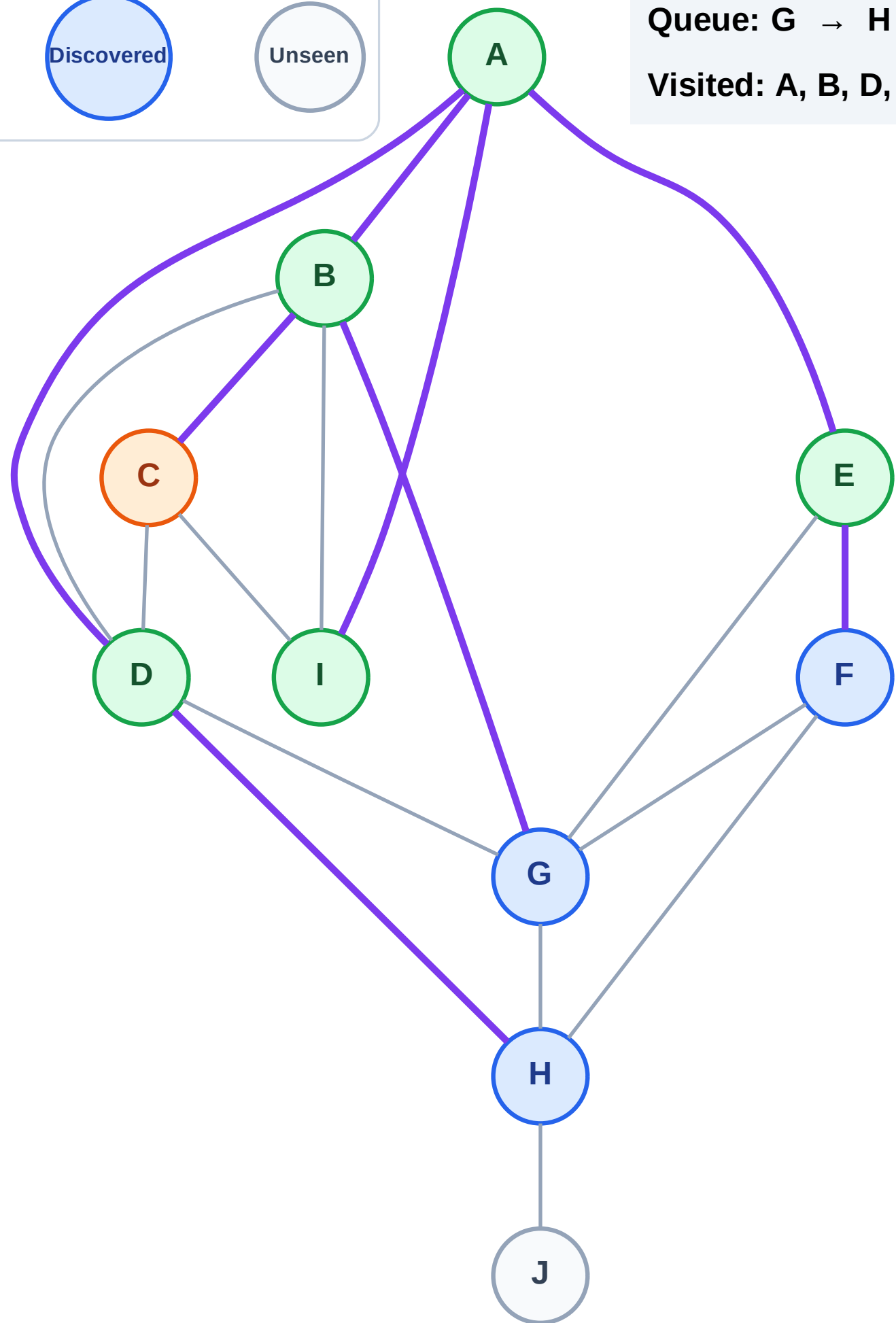


BFS Traversal

Step 12: Process node C



Queue: G → H → F
Visited: A, B, D, E, I



BFS Traversal

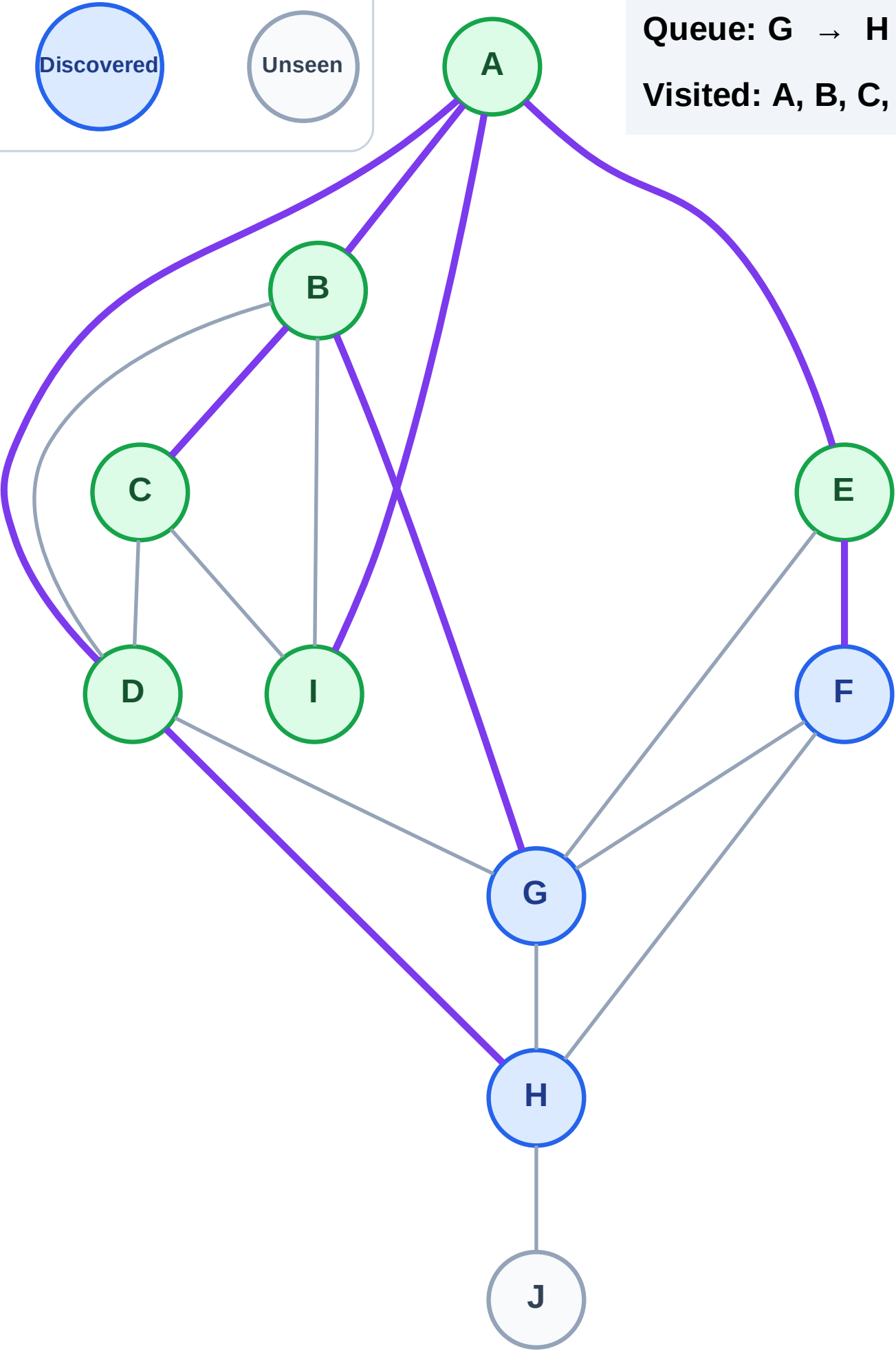
Step 13: No new neighbors from C

Legend



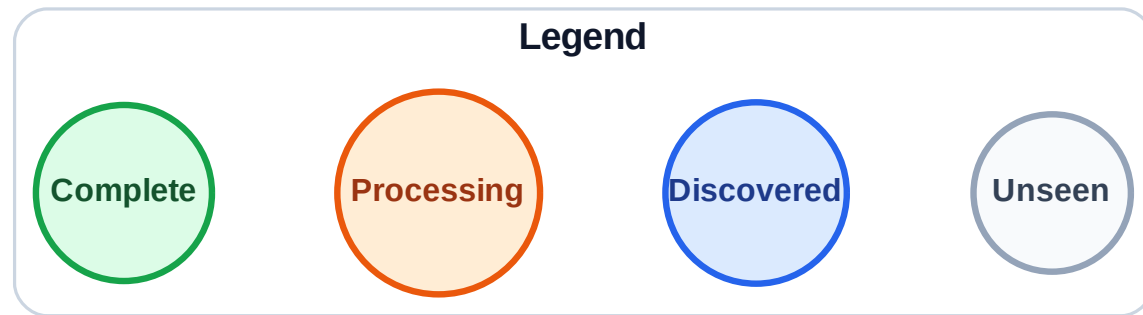
Queue: G → H → F

Visited: A, B, C, D, E, I

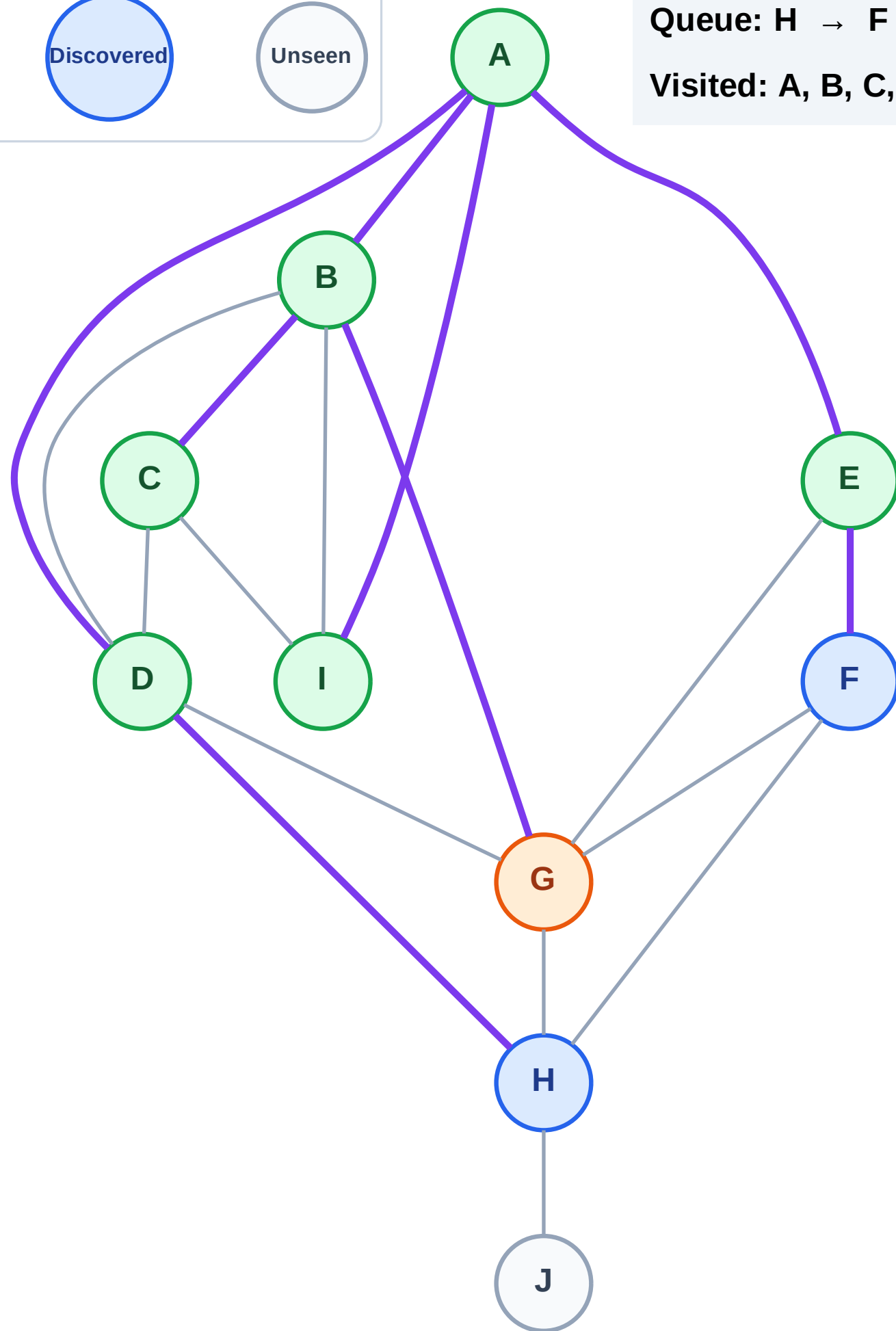


BFS Traversal

Step 14: Process node G



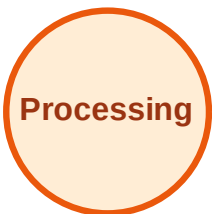
Queue: H → F
Visited: A, B, C, D, E, I



BFS Traversal

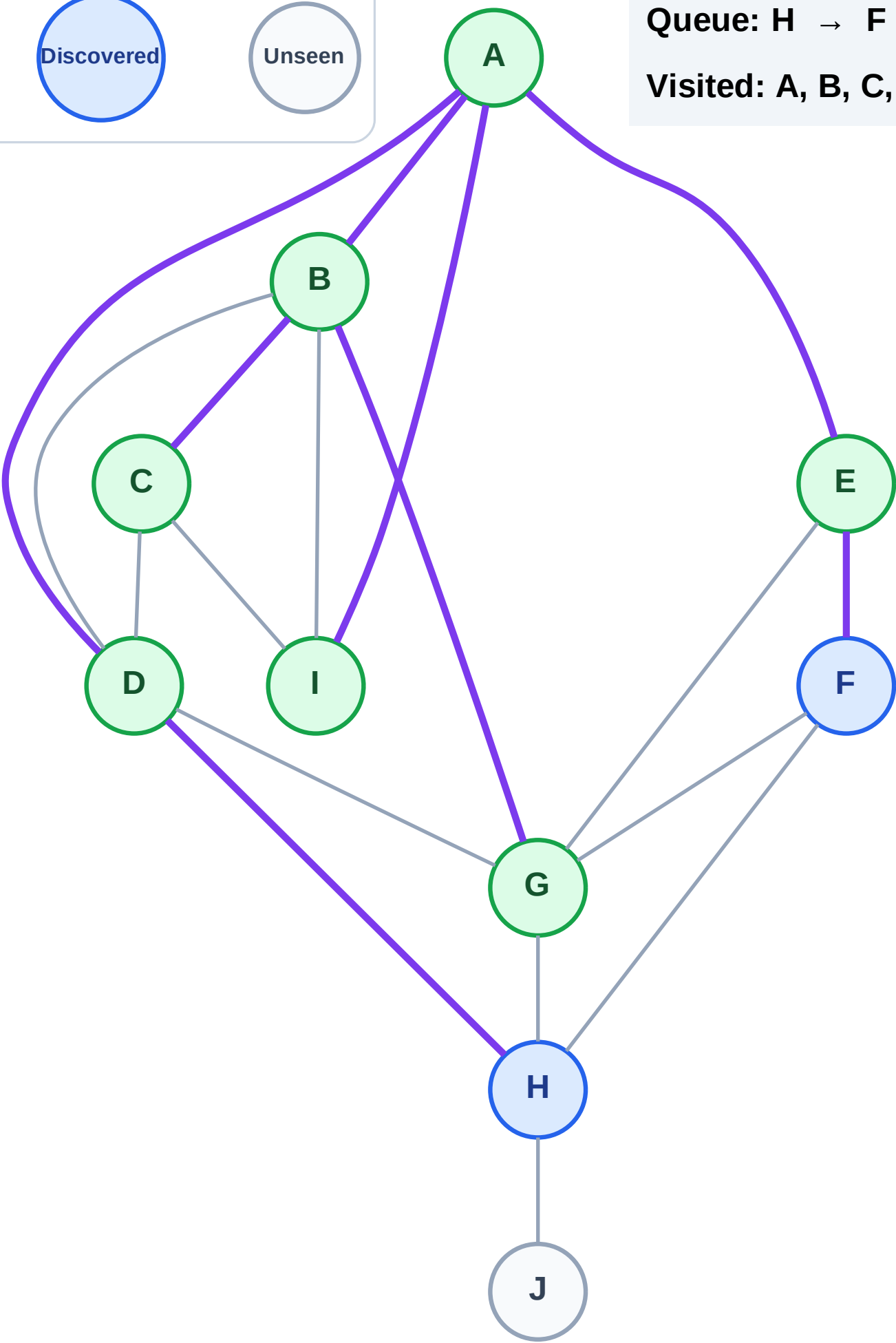
Step 15: No new neighbors from G

Legend



Queue: H → F

Visited: A, B, C, D, E, G, I



BFS Traversal

Step 16: Process node H

Legend

Complete

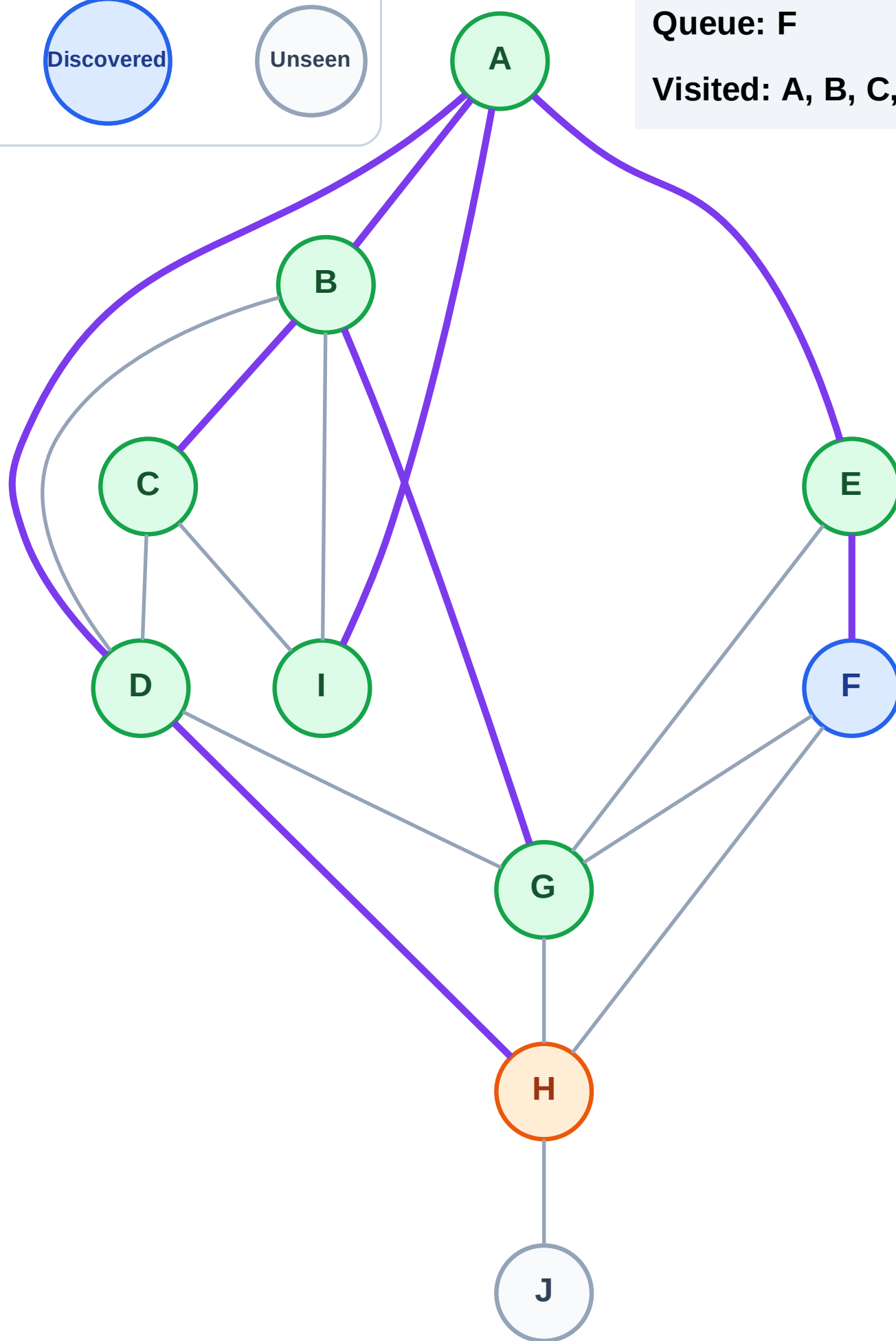
Processing

Discovered

Unseen

Queue: F

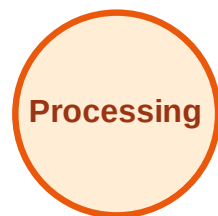
Visited: A, B, C, D, E, G, I



BFS Traversal

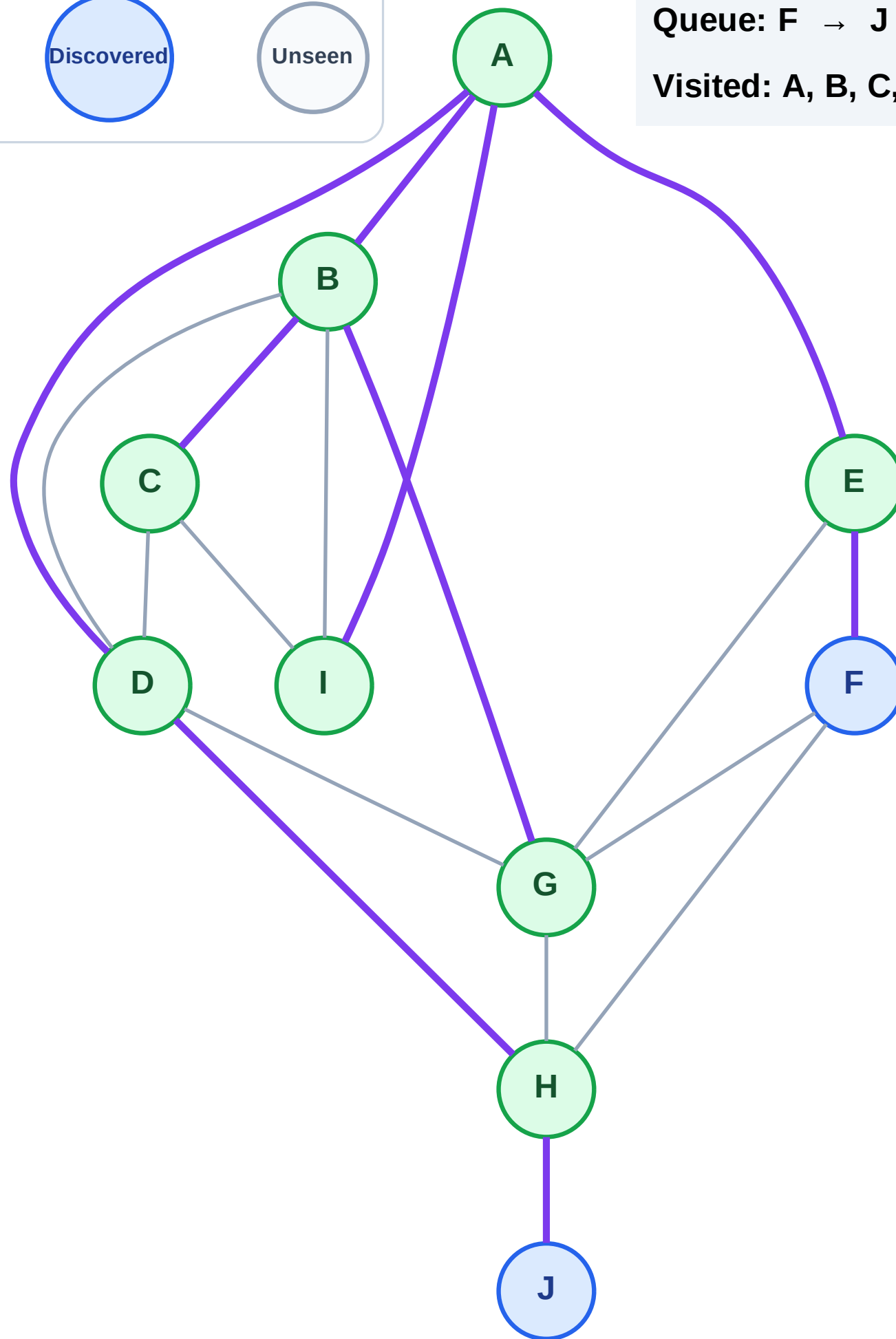
Step 17: Discover J from H

Legend



Queue: F → J

Visited: A, B, C, D, E, G, H, I



BFS Traversal

Step 18: Process node F

Legend

Complete

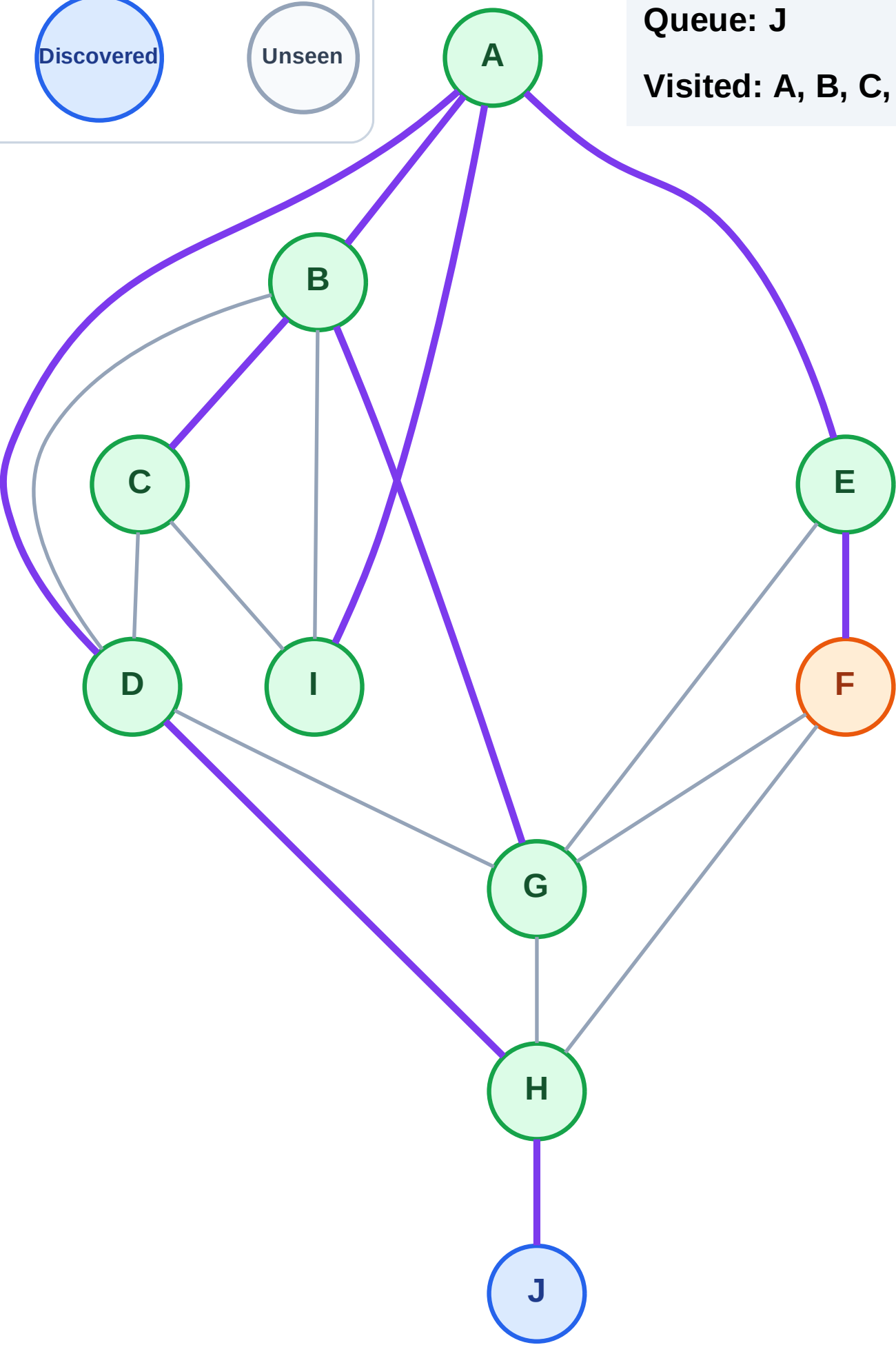
Processing

Discovered

Unseen

Queue: J

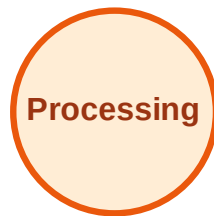
Visited: A, B, C, D, E, G, H, I



BFS Traversal

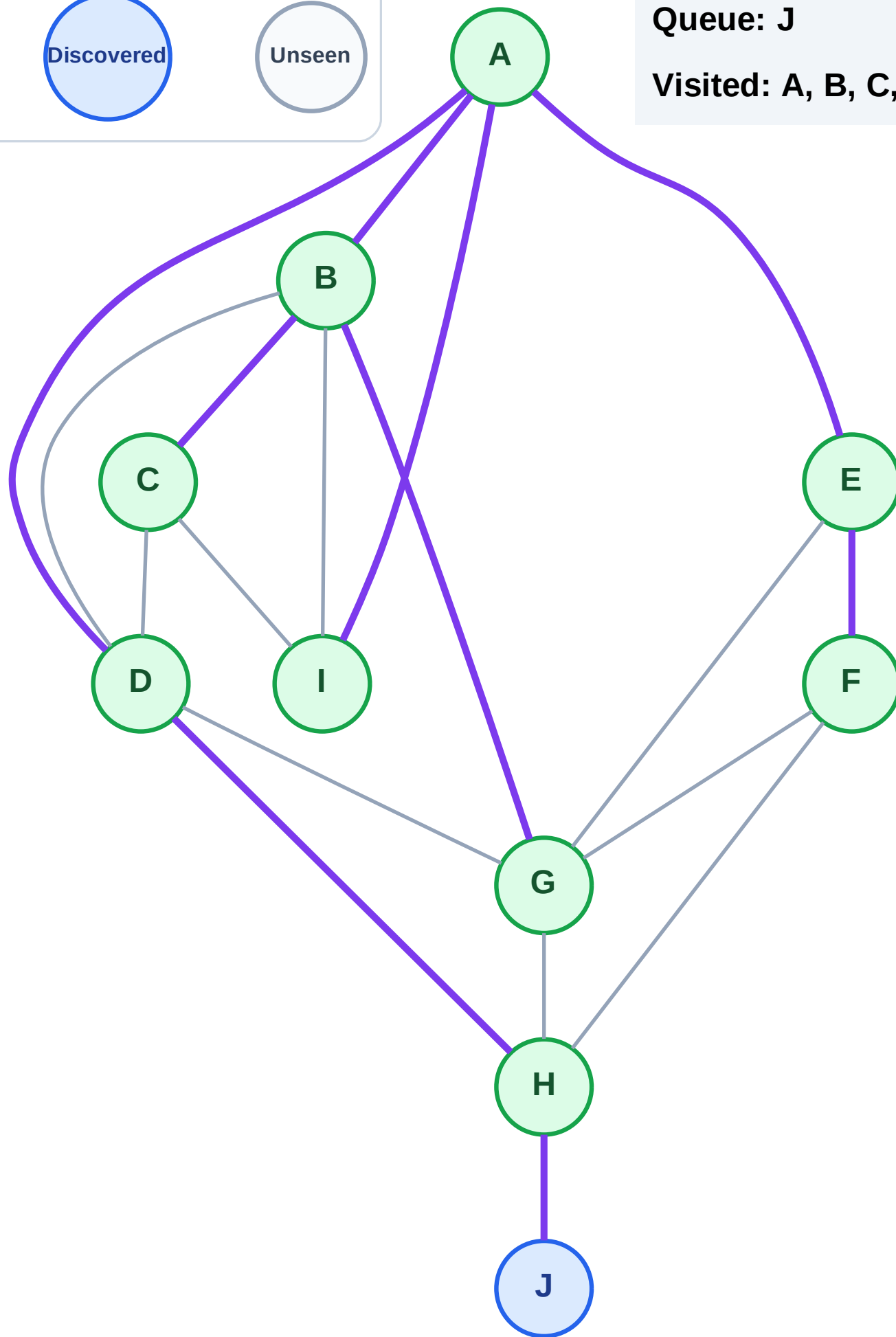
Step 19: No new neighbors from F

Legend



Queue: J

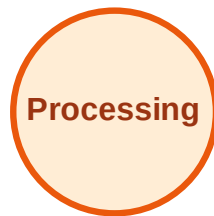
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

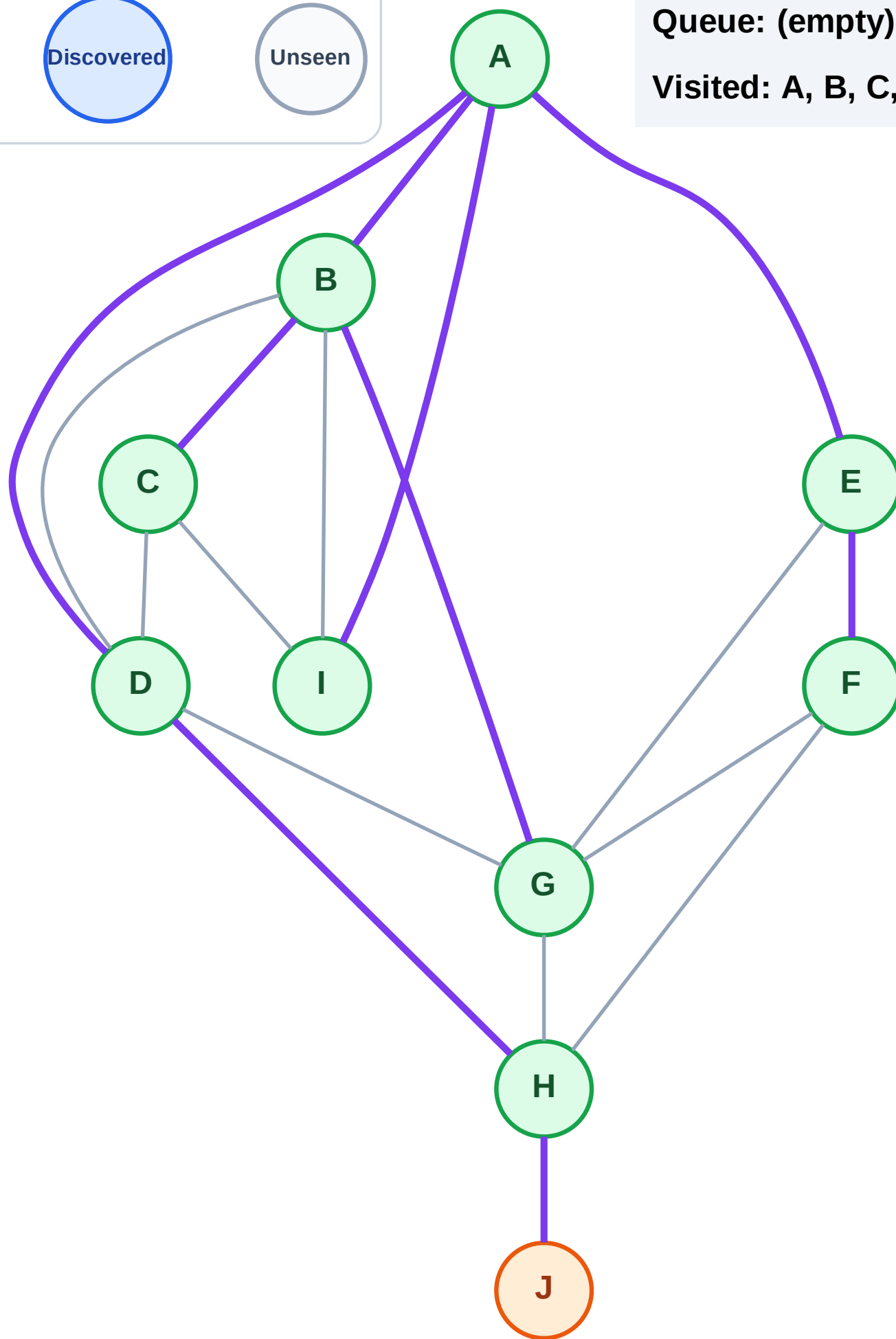
Step 20: Process node J

Legend



Queue: (empty)

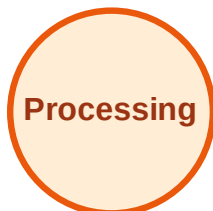
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

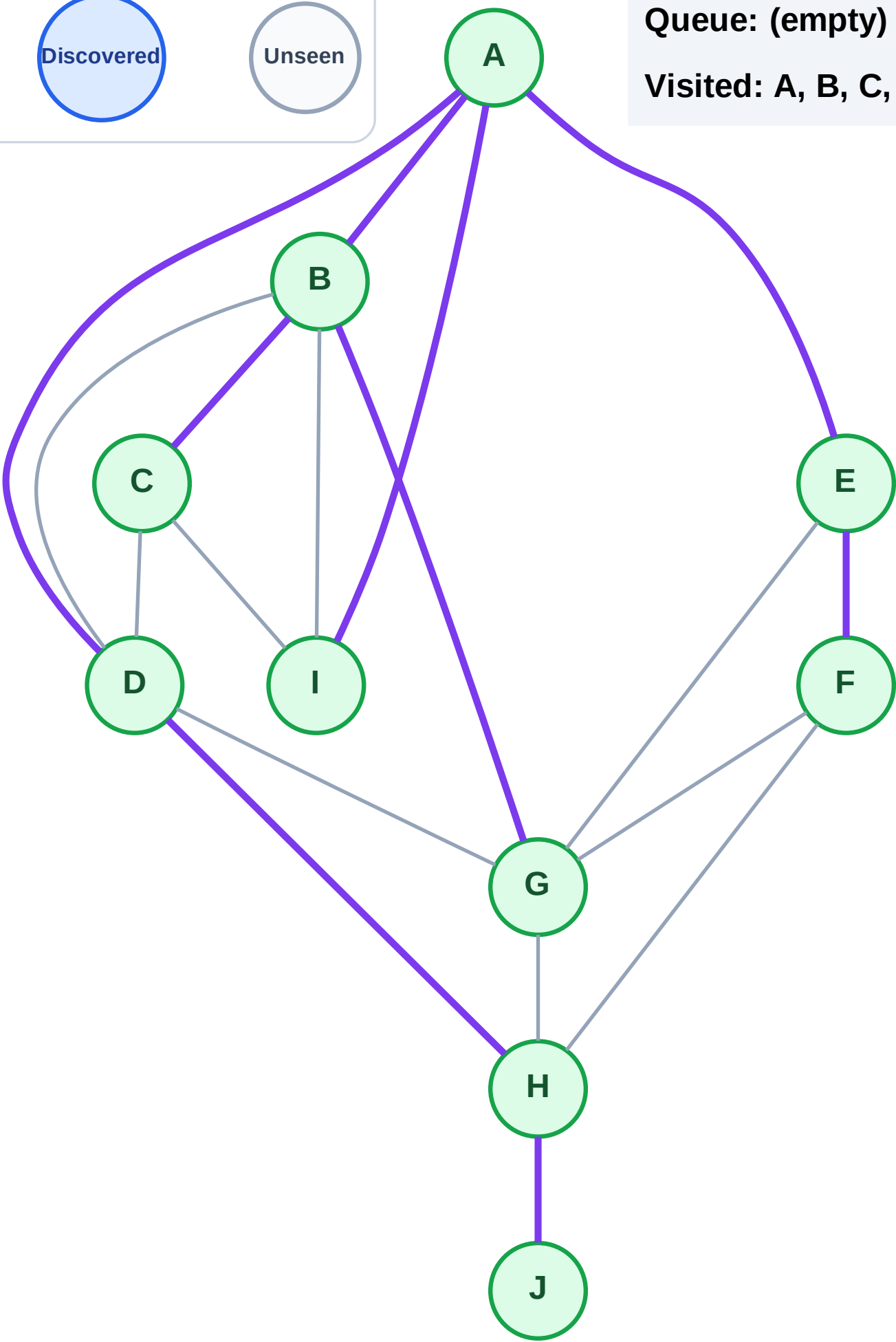
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

Legend

Complete

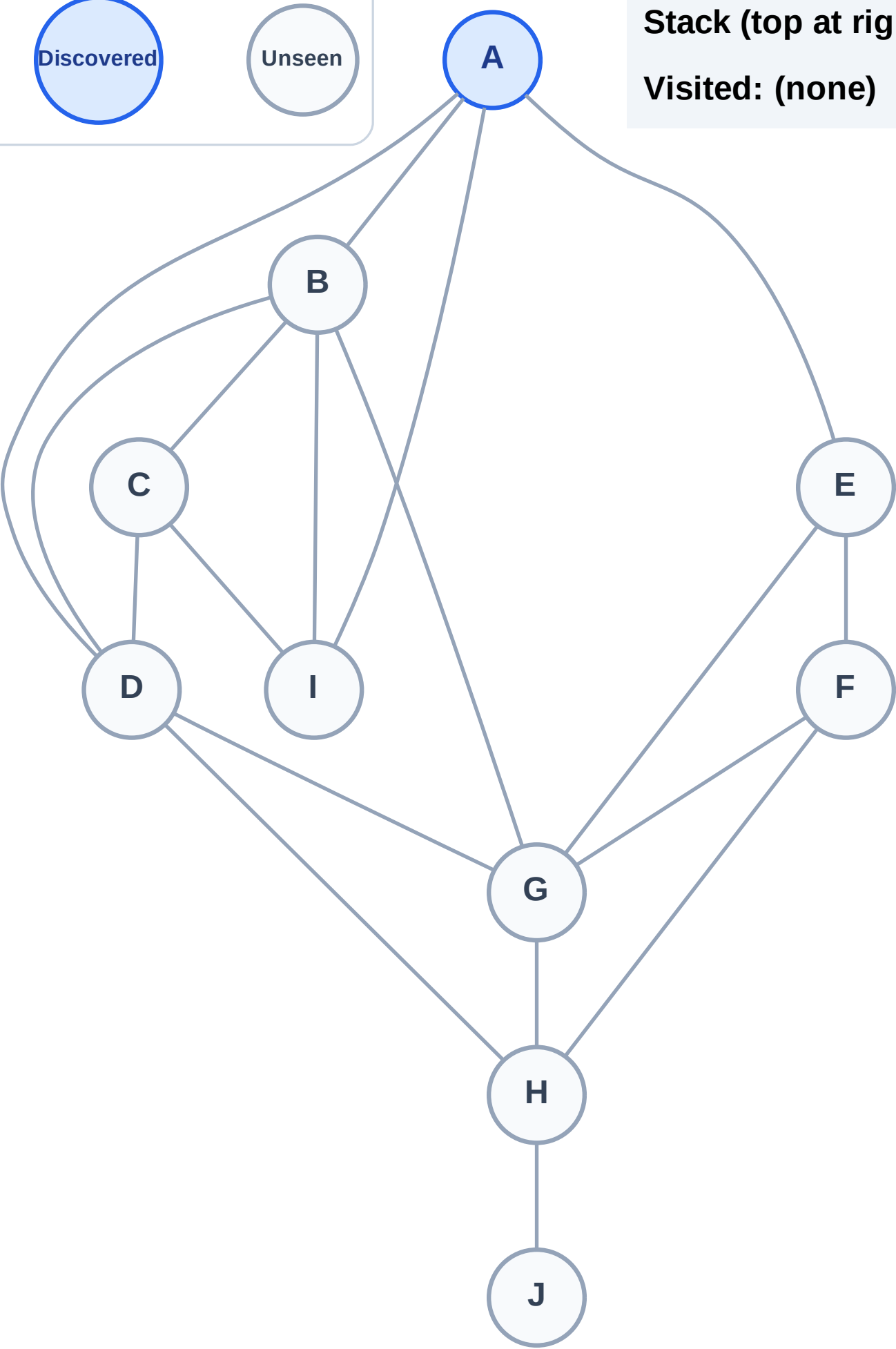
Processing

Discovered

Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

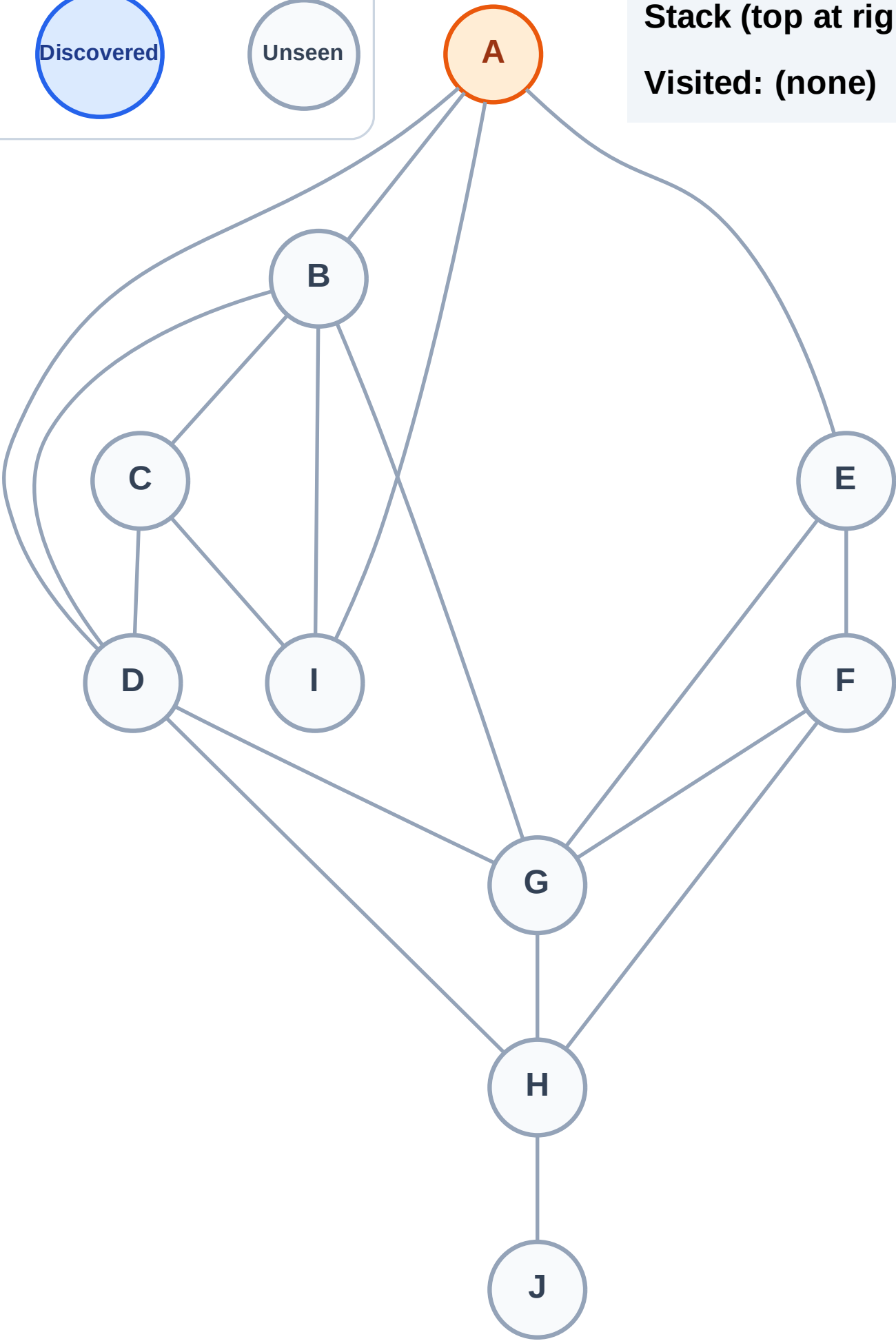
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover I, E, D, B from A

Legend

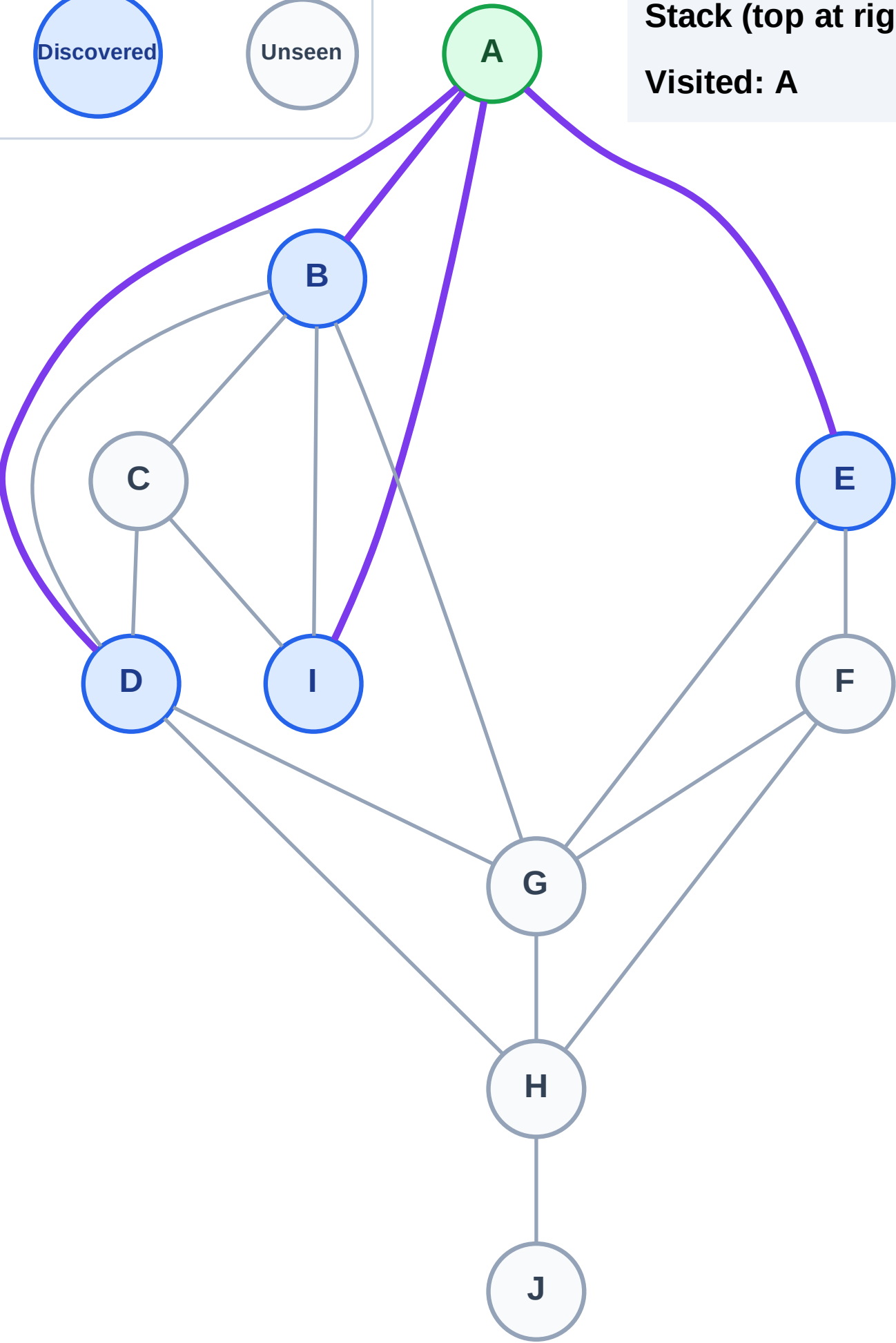
Complete

Processing

Discovered

Unseen

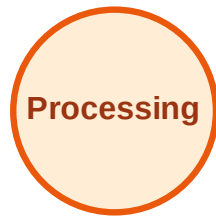
Stack (top at right): I | E | D | B
Visited: A



DFS Traversal

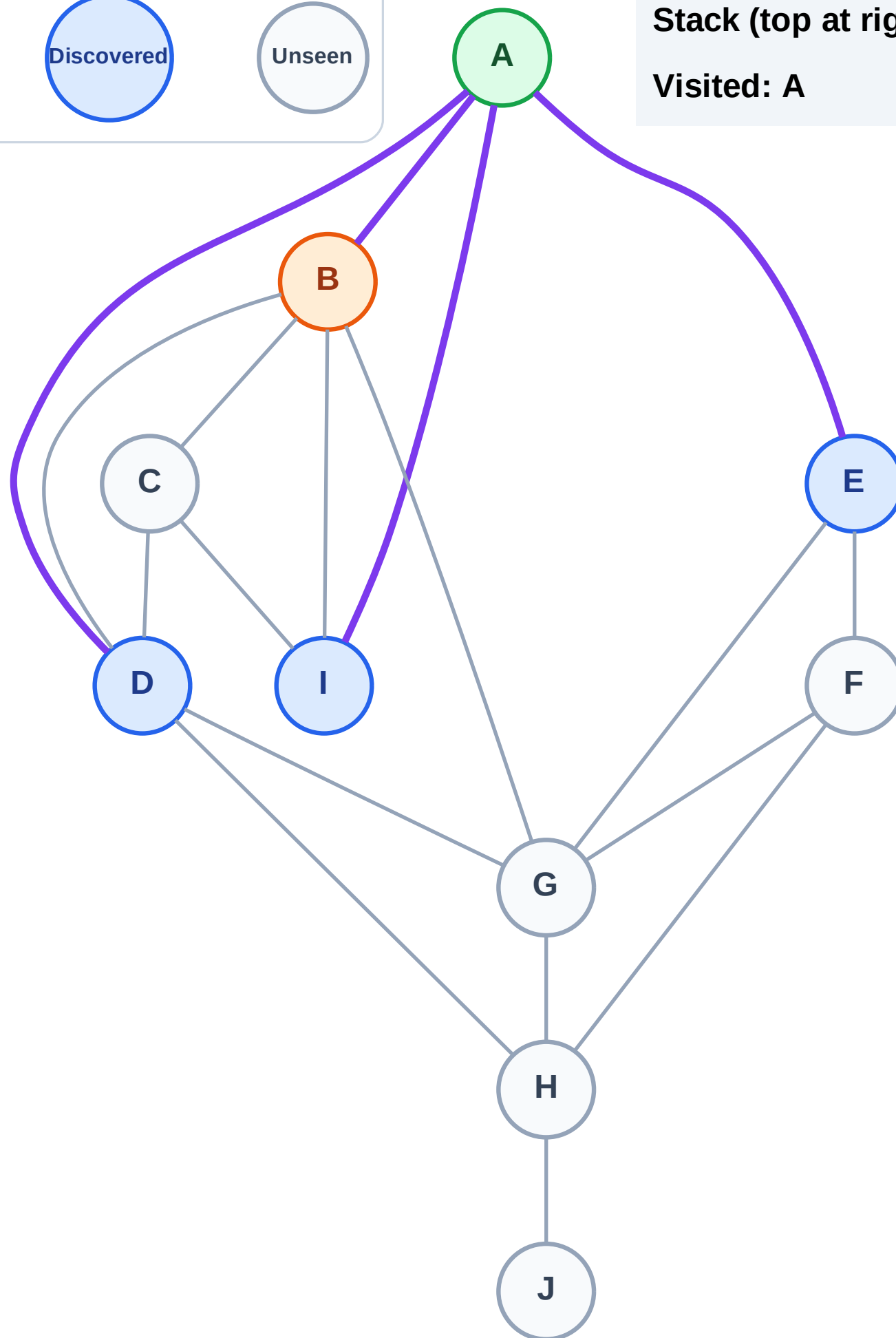
Step 4: Process node B

Legend



Stack (top at right): I | E | D

Visited: A



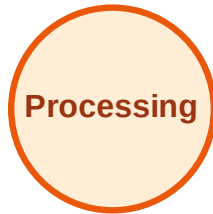
DFS Traversal

Step 5: Discover G, C from B

Legend



Complete



Processing



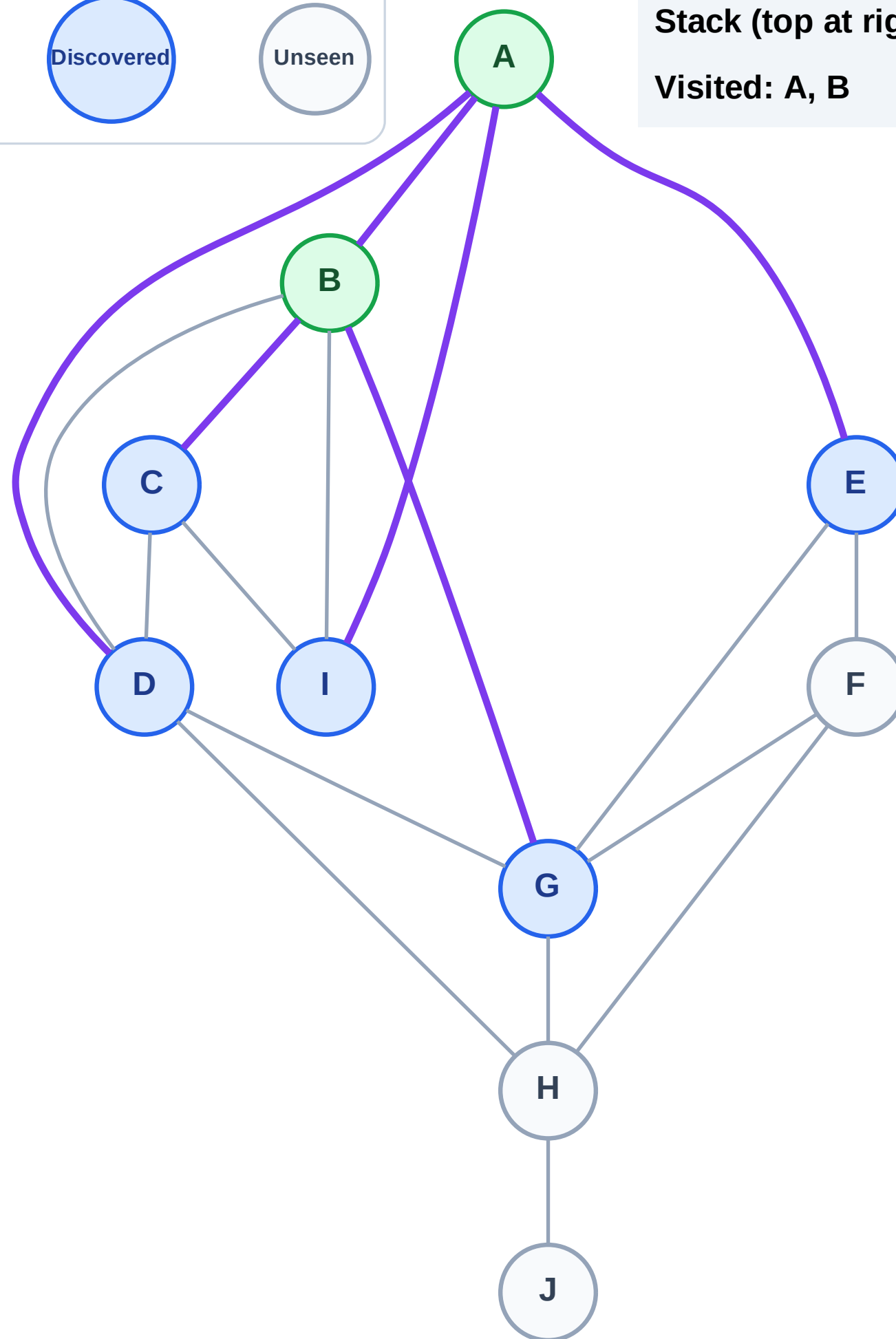
Discovered



Unseen

Stack (top at right): I | E | D | G | C

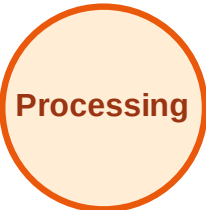
Visited: A, B



DFS Traversal

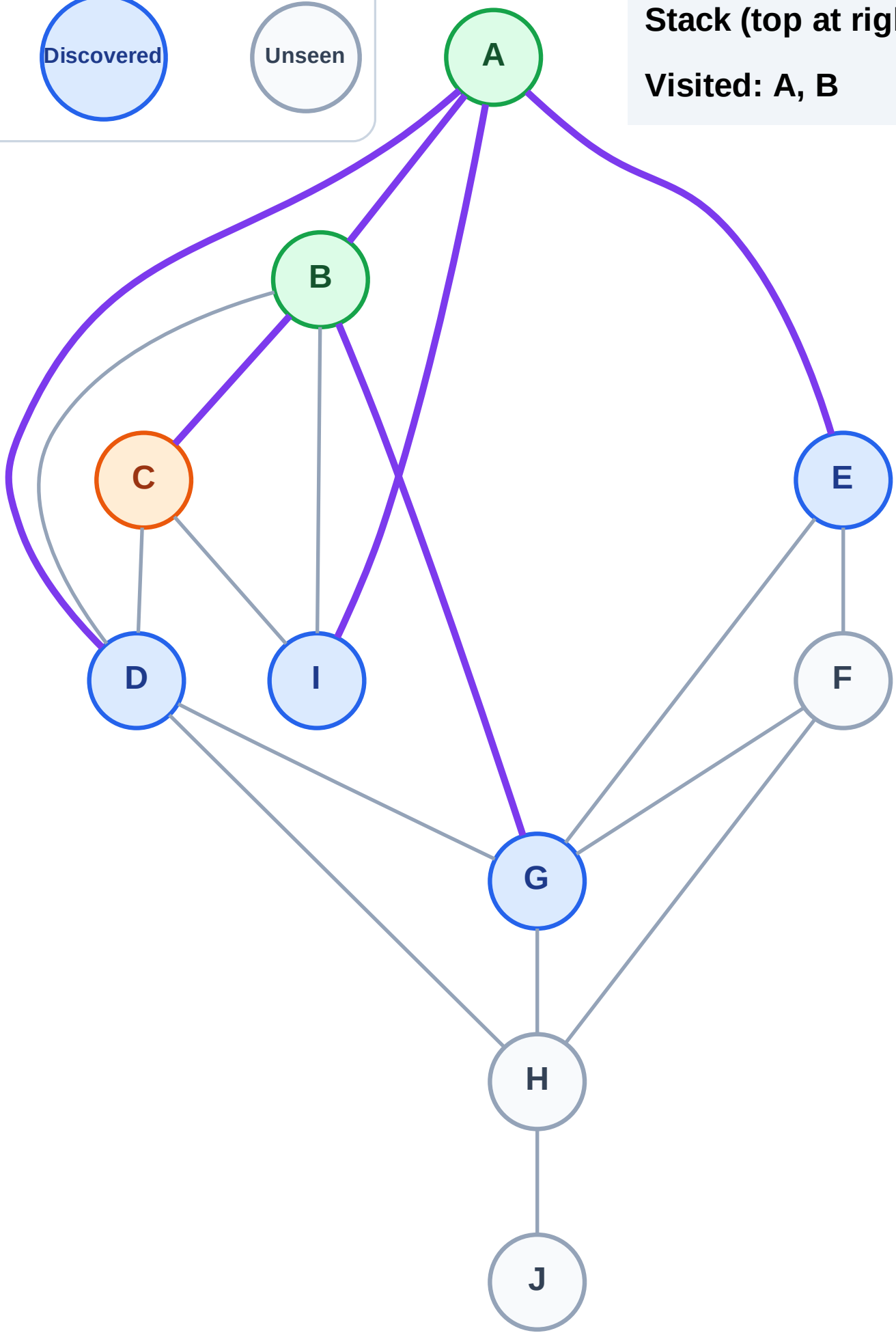
Step 6: Process node C

Legend



Stack (top at right): I | E | D | G

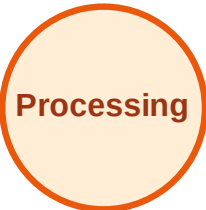
Visited: A, B



DFS Traversal

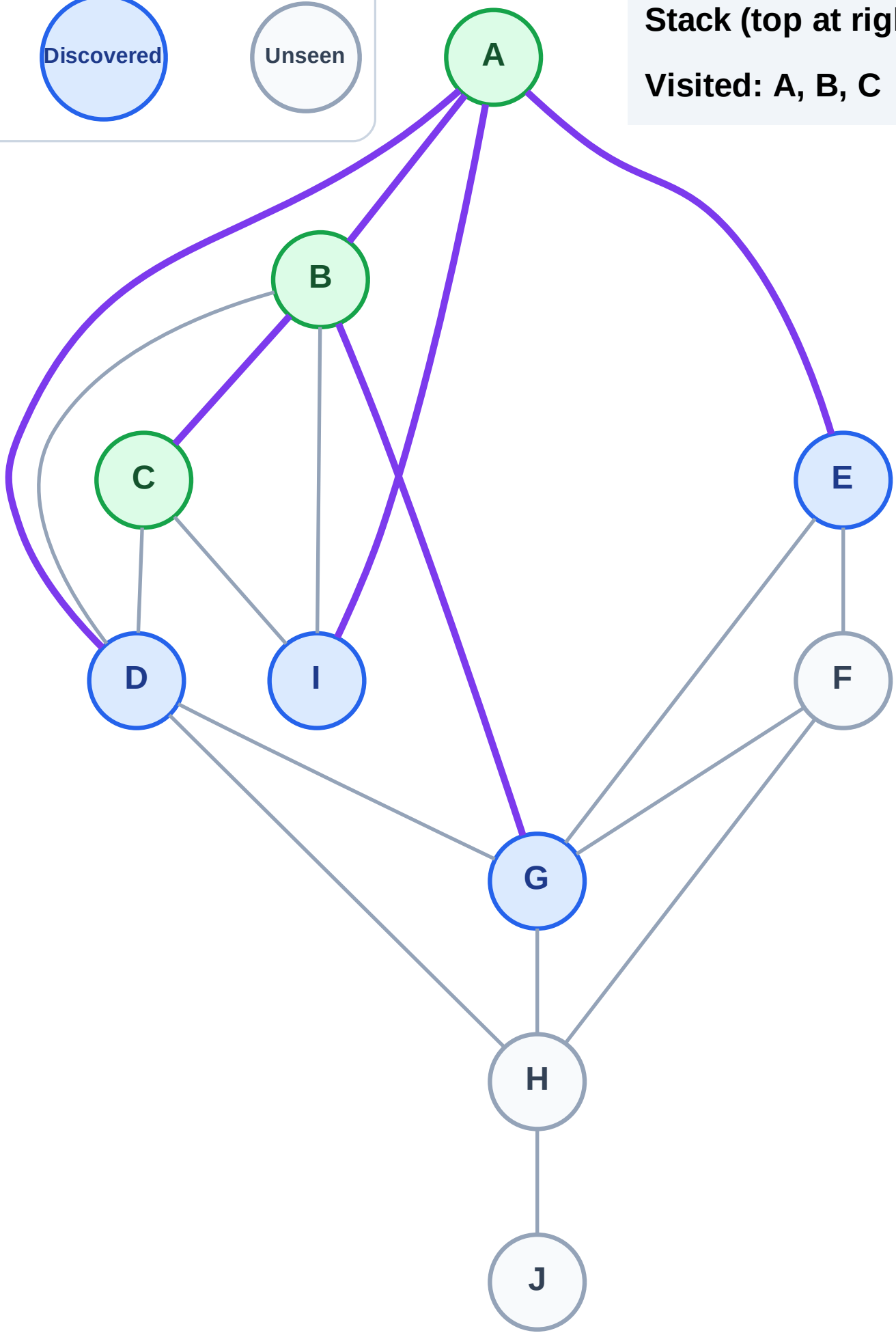
Step 7: No new neighbors from C

Legend



Stack (top at right): I | E | D | G

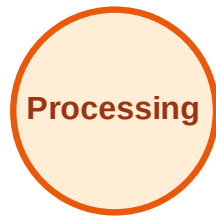
Visited: A, B, C



DFS Traversal

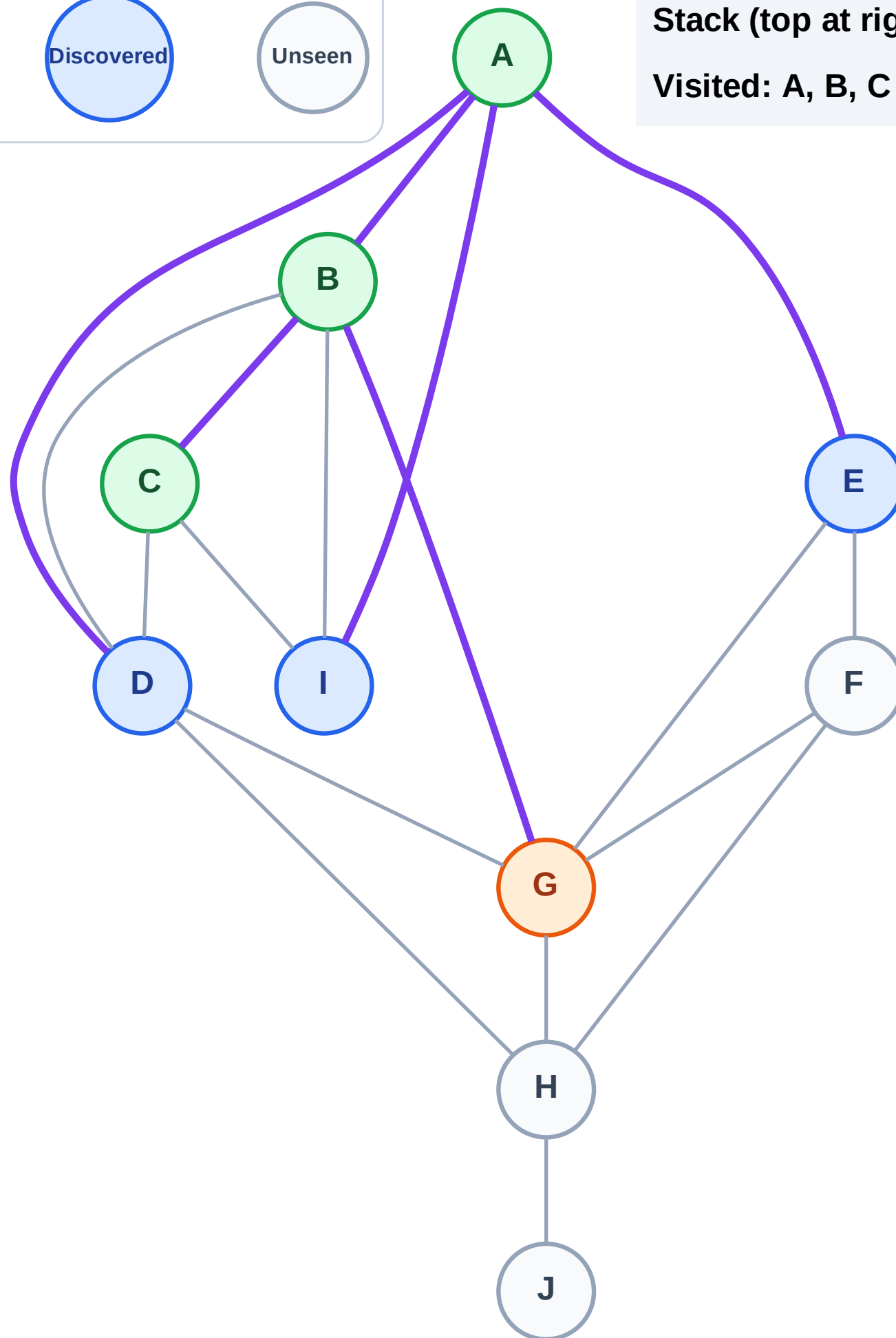
Step 8: Process node G

Legend



Stack (top at right): I | E | D

Visited: A, B, C



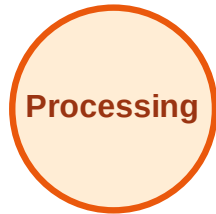
DFS Traversal

Step 9: Discover H, F from G

Legend



Complete



Processing



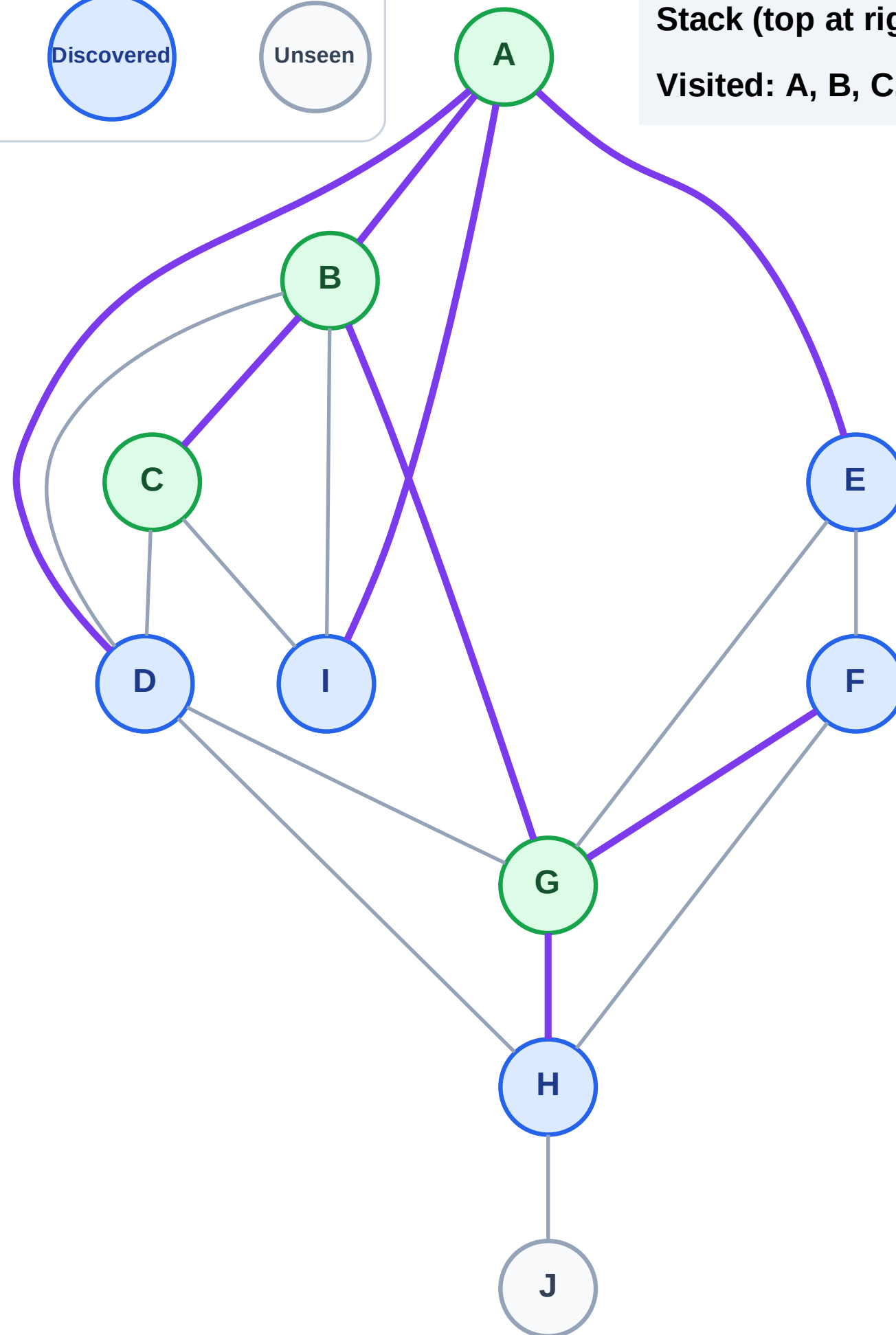
Discovered



Unseen

Stack (top at right): I | E | D | H | F

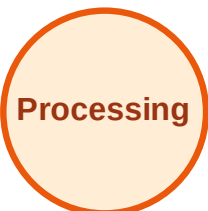
Visited: A, B, C, G



DFS Traversal

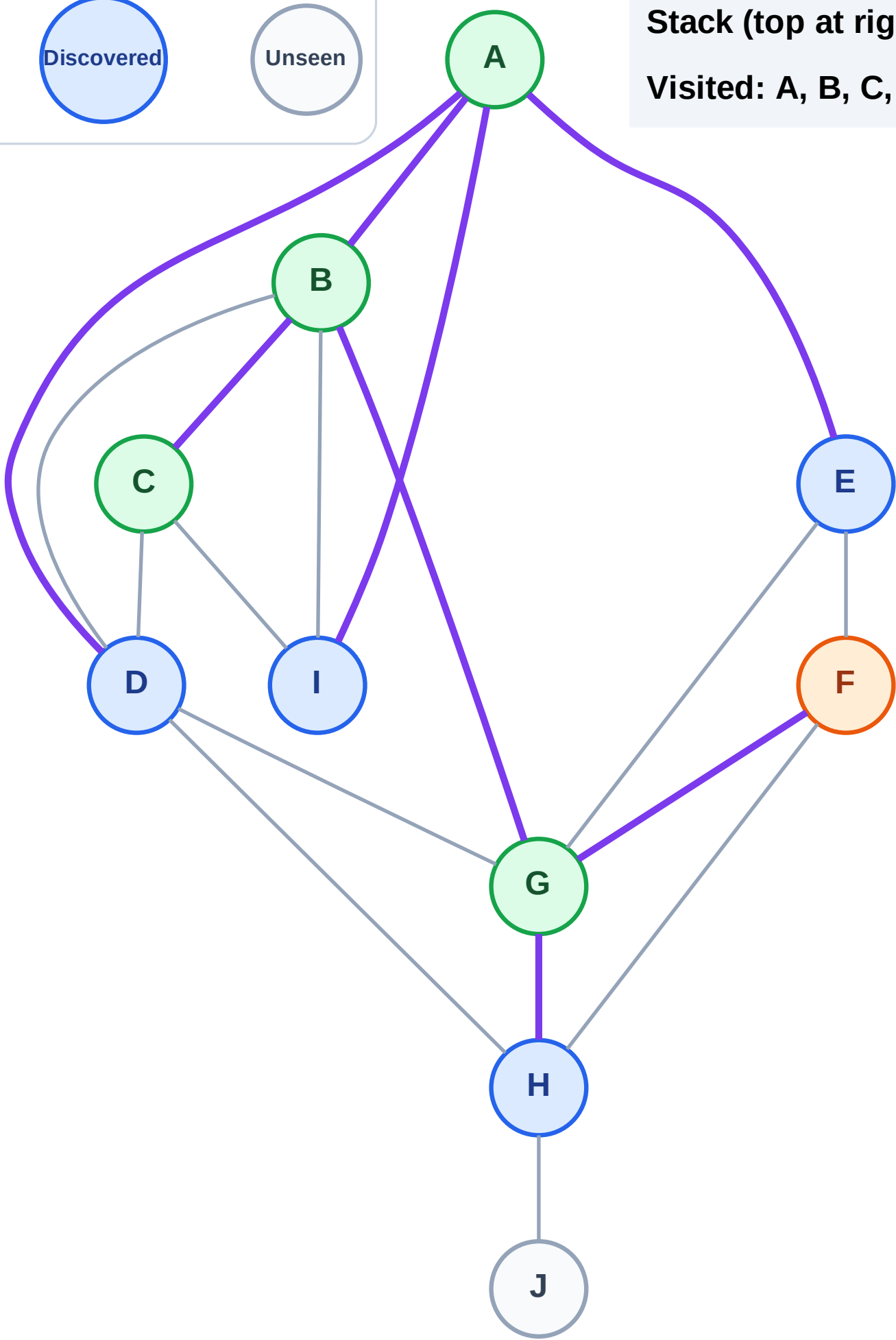
Step 10: Process node F

Legend



Stack (top at right): I | E | D | H

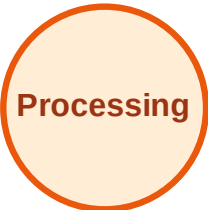
Visited: A, B, C, G



DFS Traversal

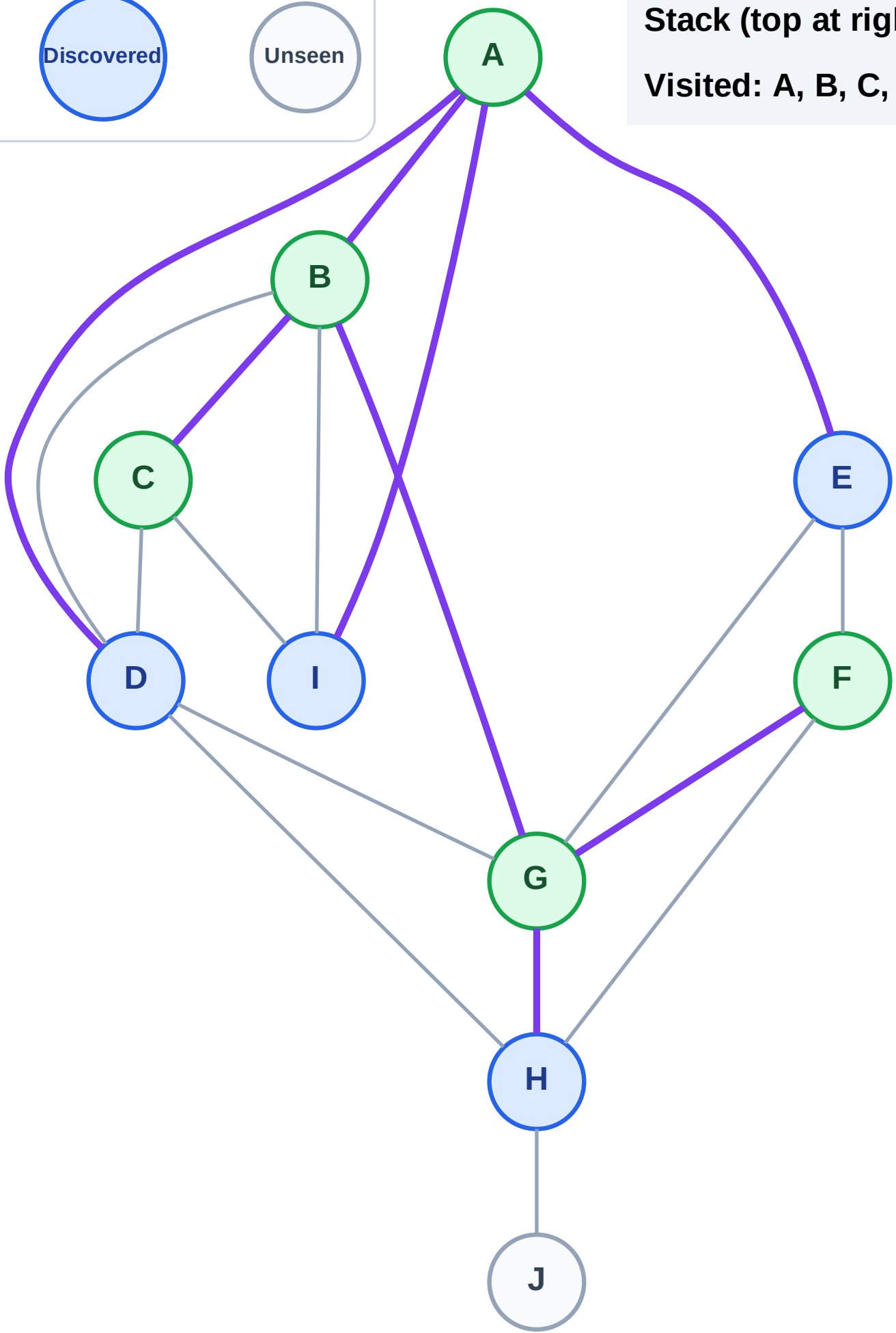
Step 11: No new neighbors from F

Legend



Stack (top at right): I | E | D | H

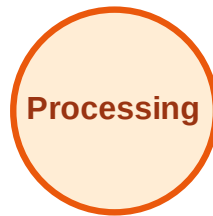
Visited: A, B, C, F, G



DFS Traversal

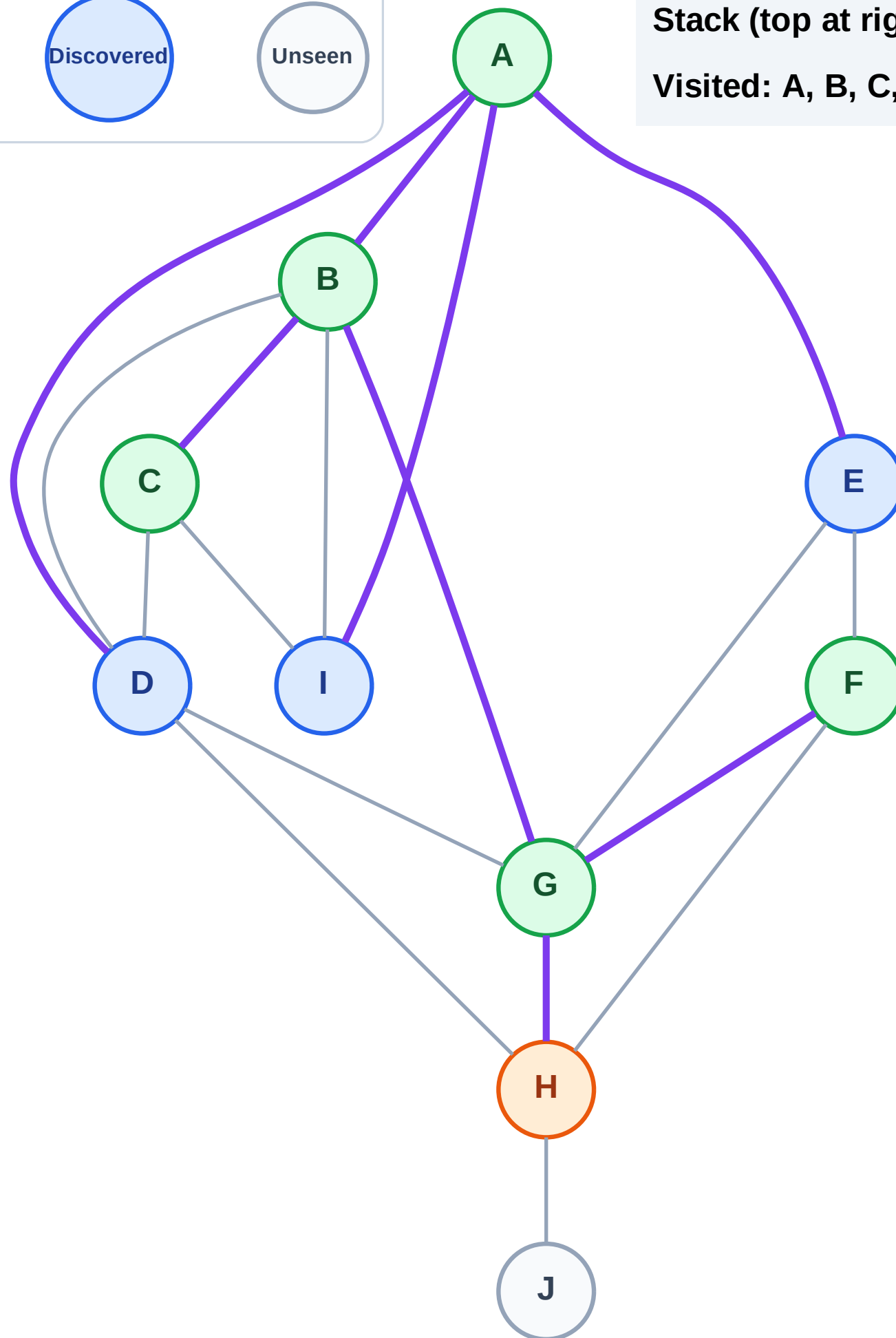
Step 12: Process node H

Legend



Stack (top at right): I | E | D

Visited: A, B, C, F, G



DFS Traversal

Step 13: Discover J from H

Legend

Complete

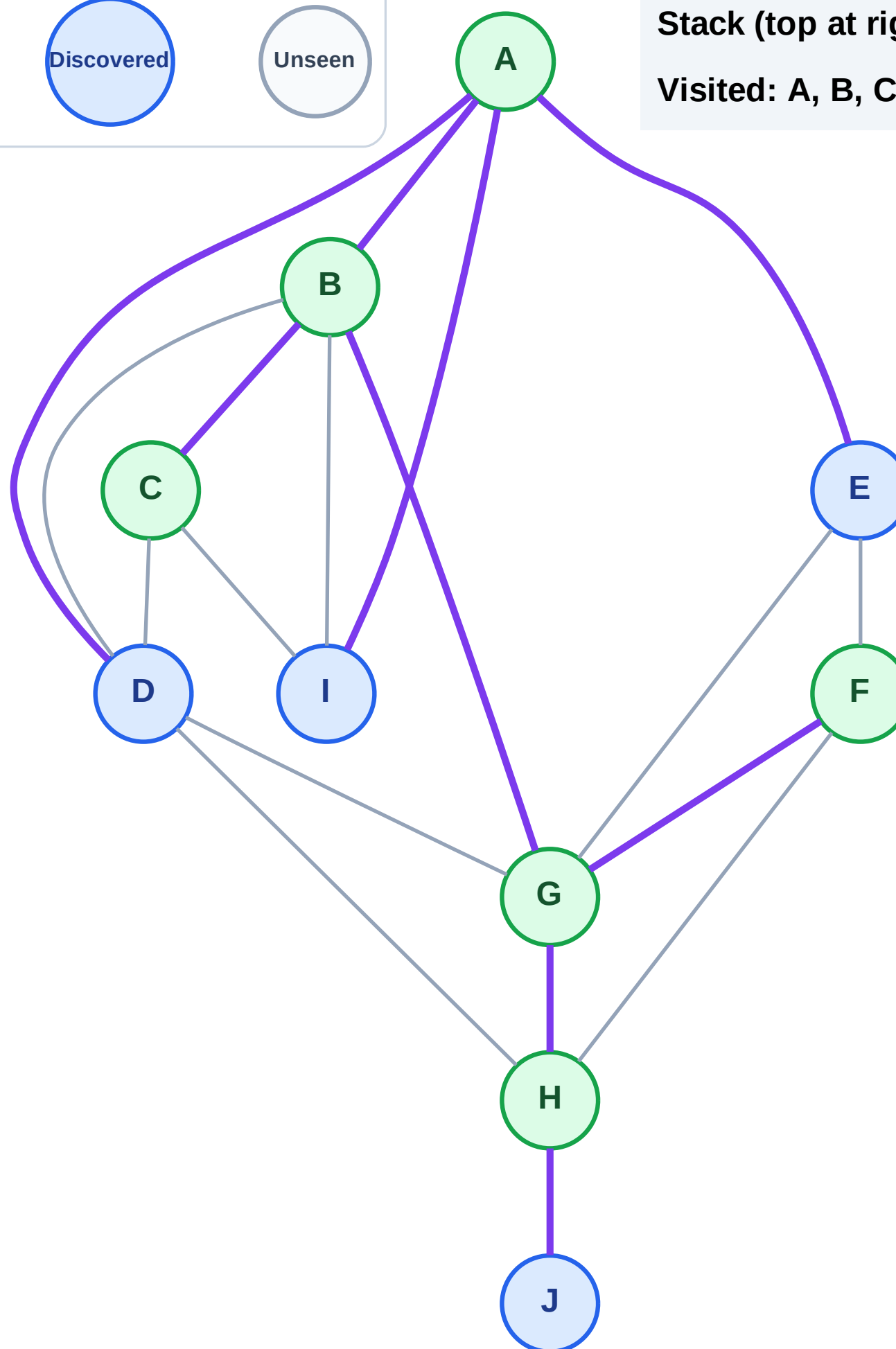
Processing

Discovered

Unseen

Stack (top at right): I | E | D | J

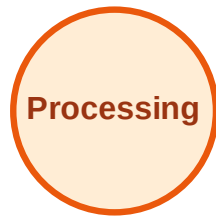
Visited: A, B, C, F, G, H



DFS Traversal

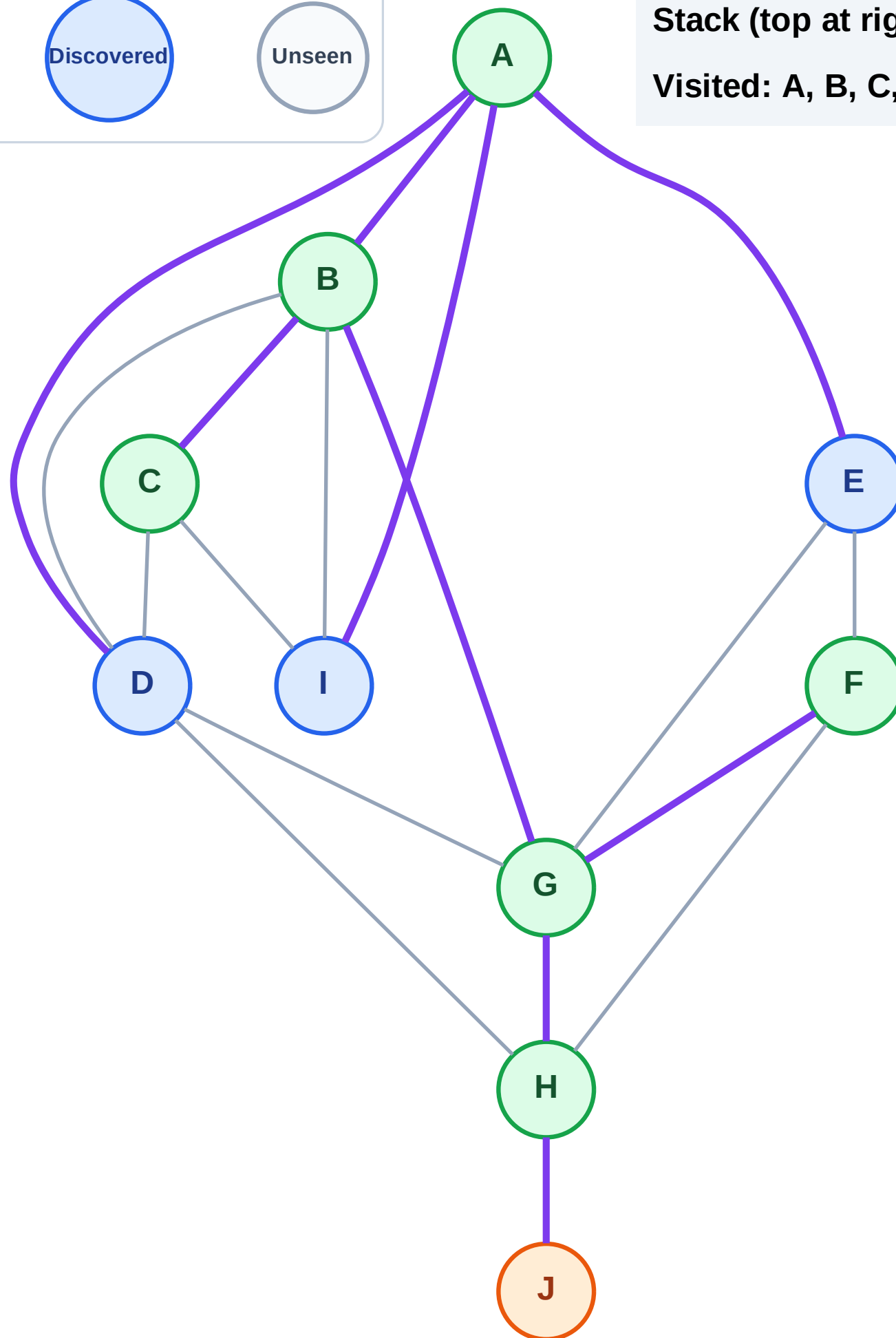
Step 14: Process node J

Legend



Stack (top at right): I | E | D

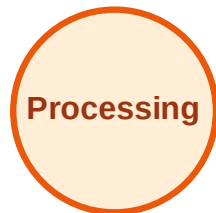
Visited: A, B, C, F, G, H



DFS Traversal

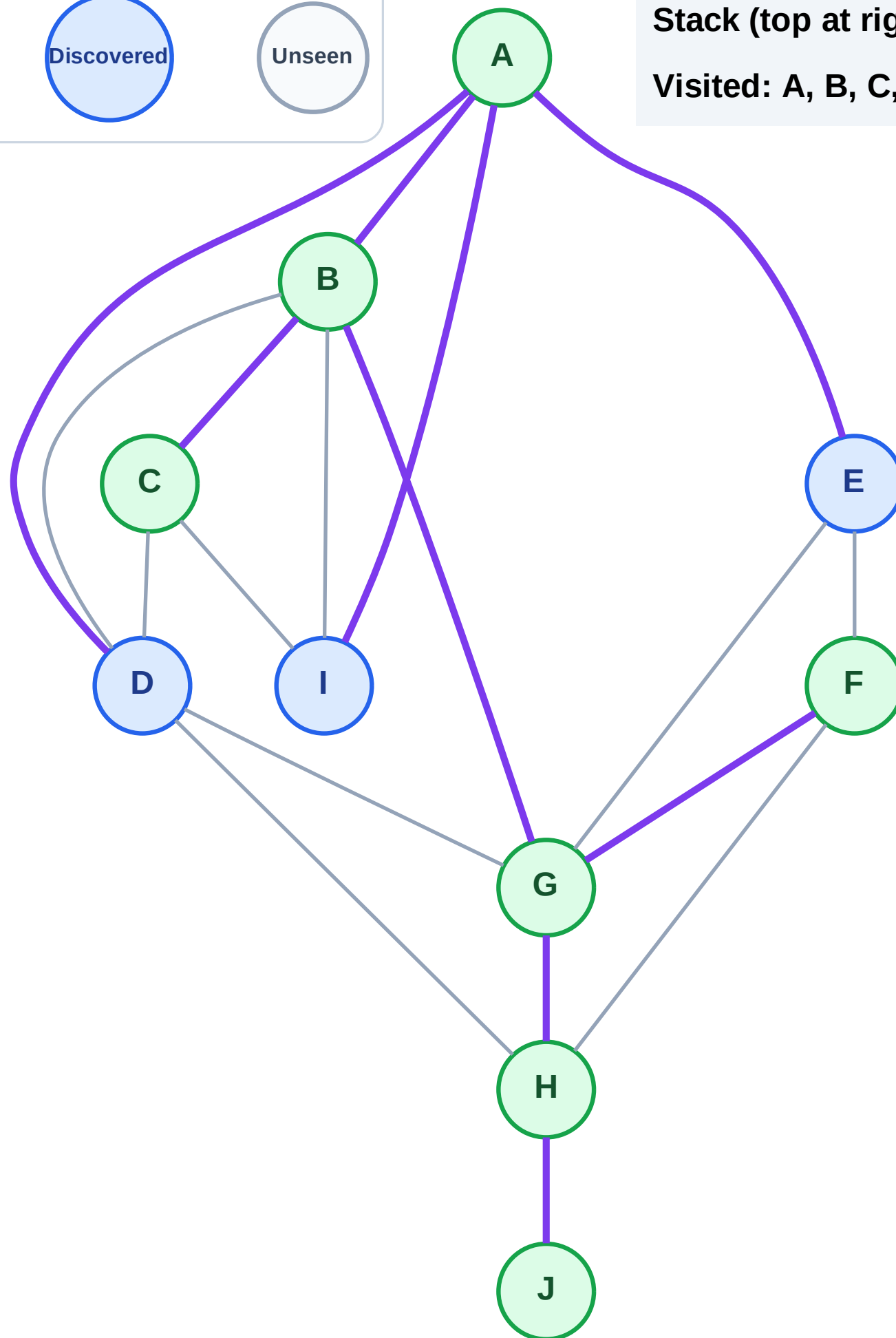
Step 15: No new neighbors from J

Legend



Stack (top at right): I | E | D

Visited: A, B, C, F, G, H, J



DFS Traversal

Step 16: Process node D

Legend

Complete

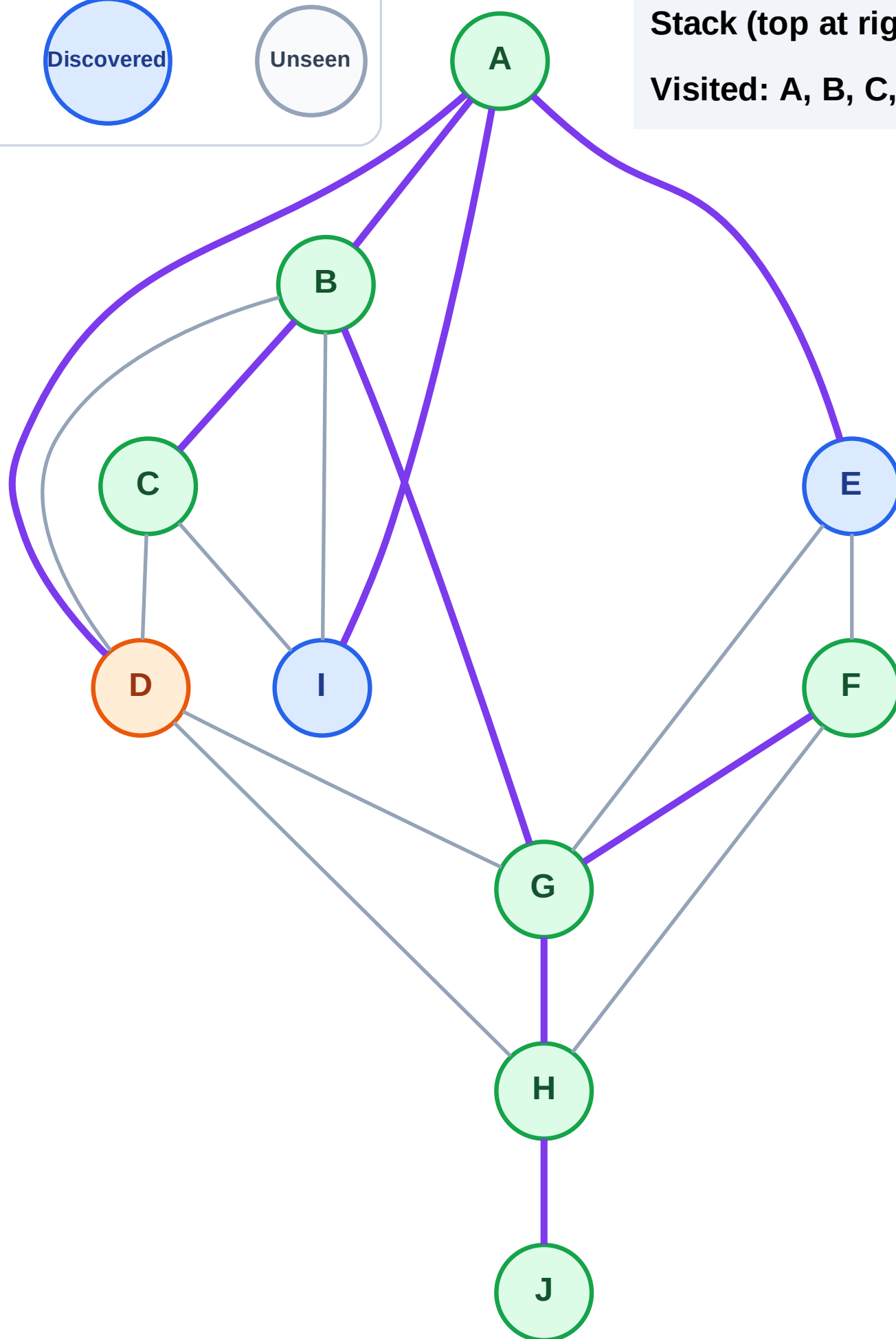
Processing

Discovered

Unseen

Stack (top at right): I | E

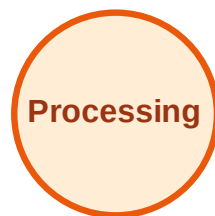
Visited: A, B, C, F, G, H, J



DFS Traversal

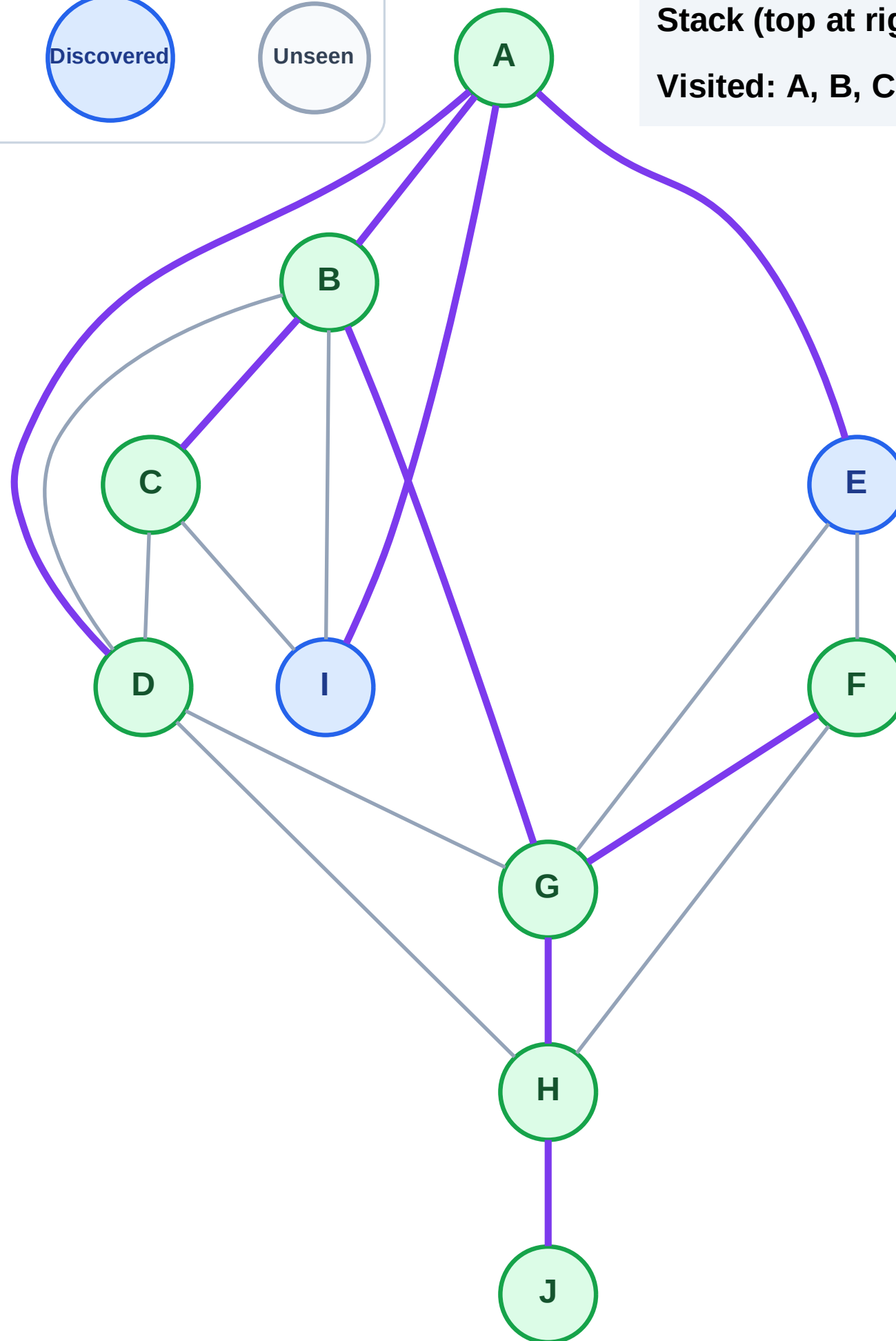
Step 17: No new neighbors from D

Legend



Stack (top at right): I | E

Visited: A, B, C, D, F, G, H, J



DFS Traversal

Step 18: Process node E

Legend

Complete

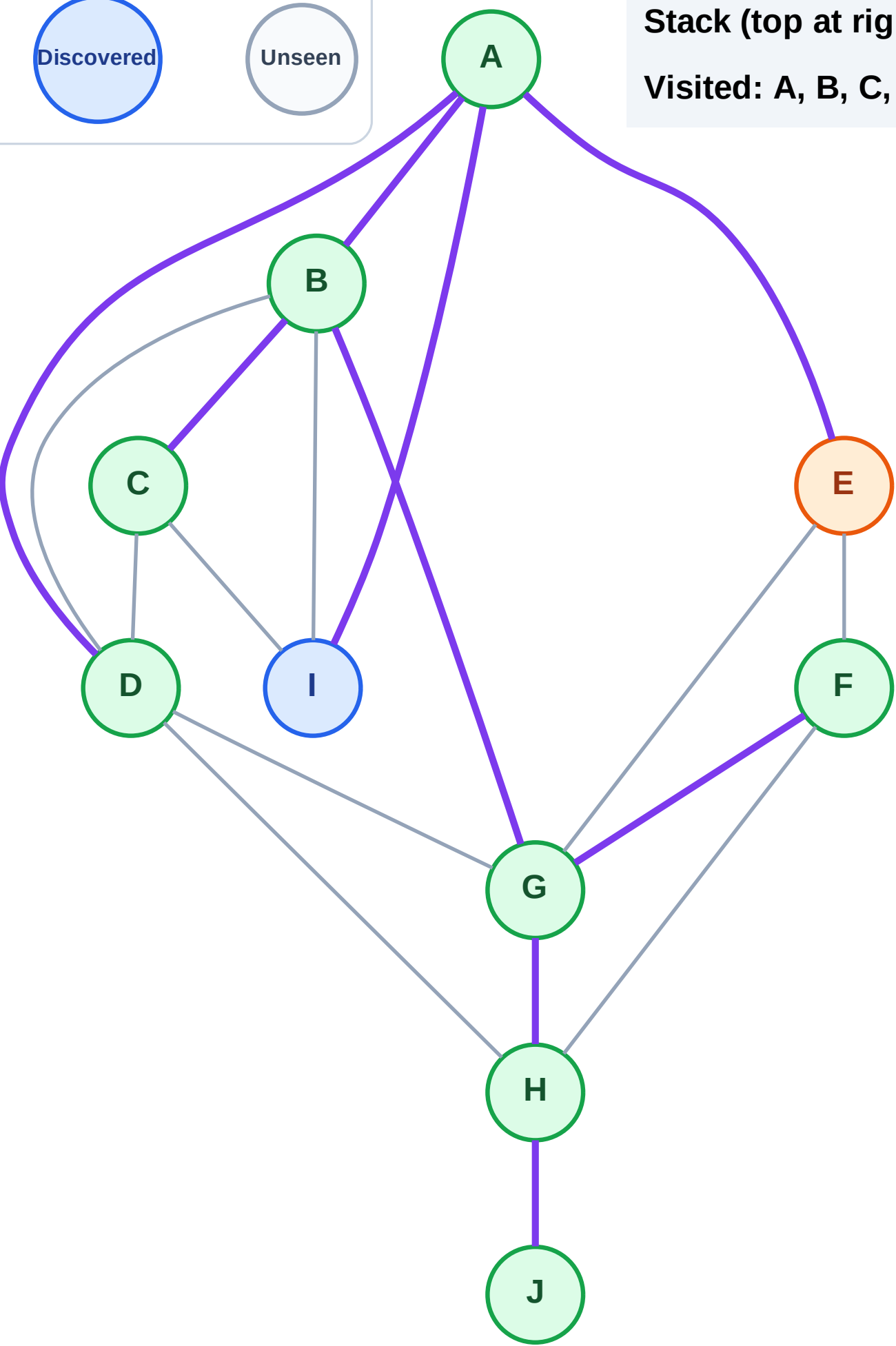
Processing

Discovered

Unseen

Stack (top at right): I

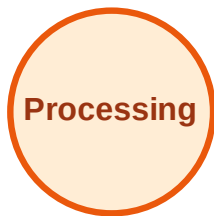
Visited: A, B, C, D, F, G, H, J



DFS Traversal

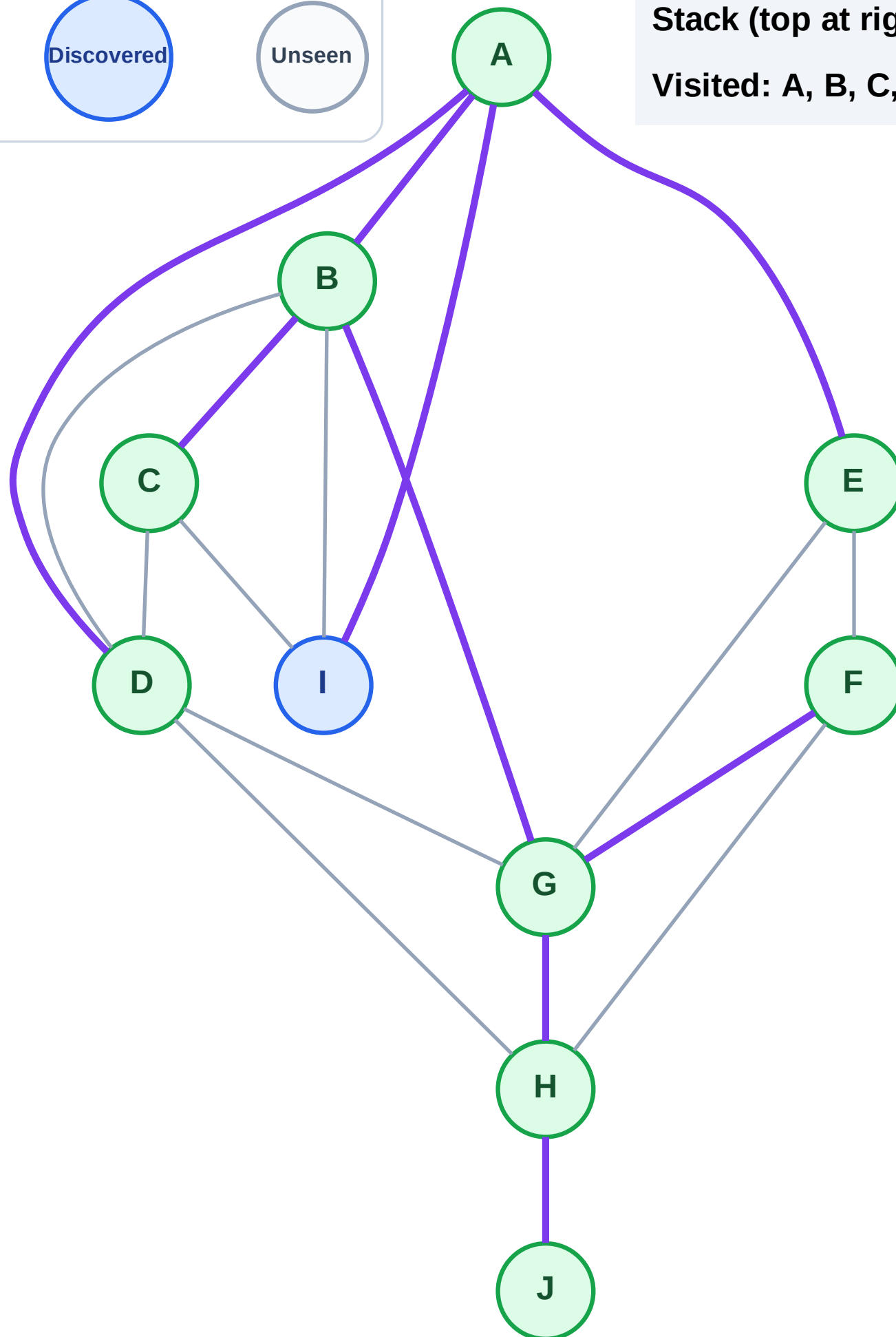
Step 19: No new neighbors from E

Legend



Stack (top at right): I

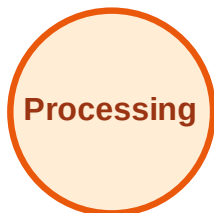
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

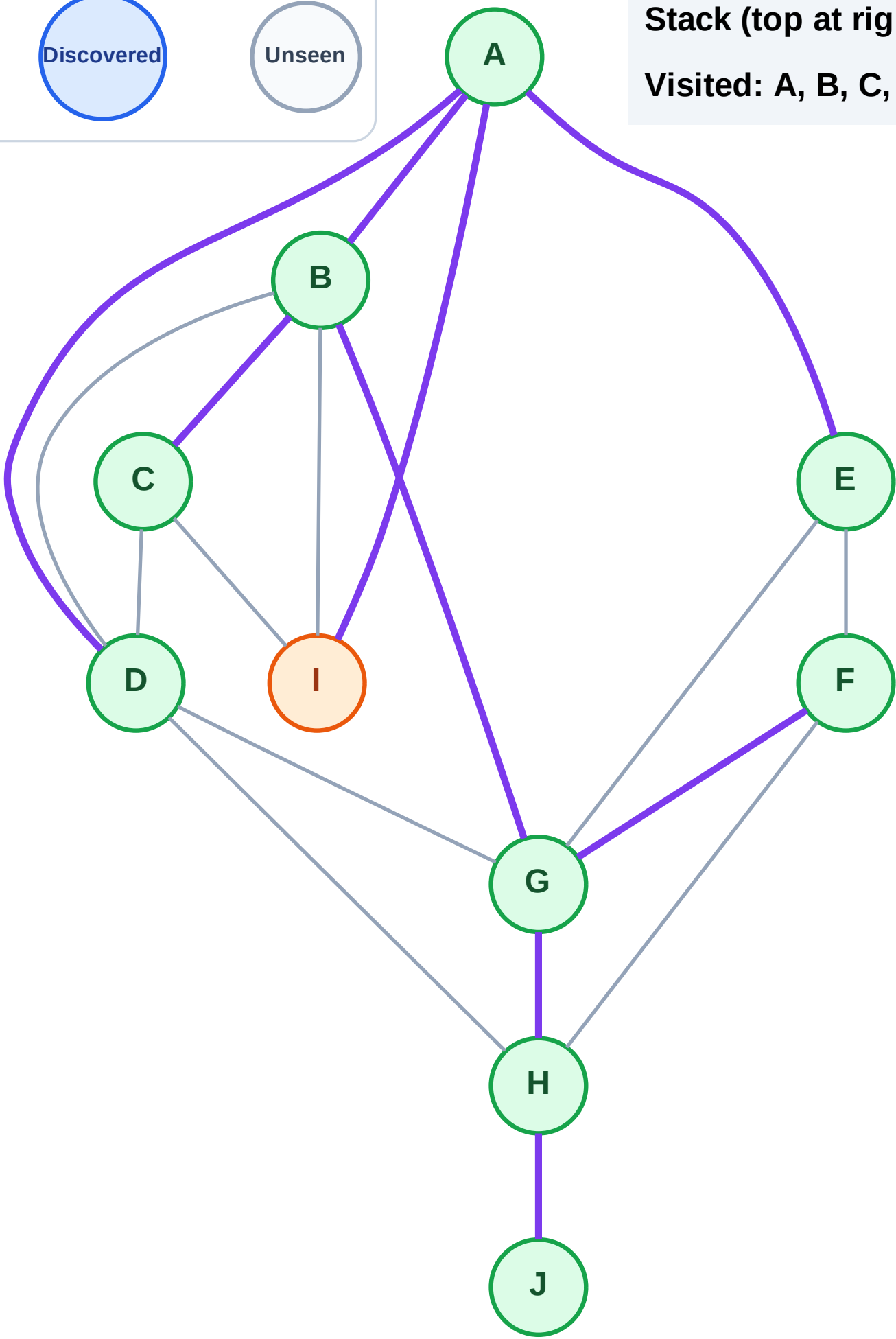
Step 20: Process node I

Legend



Stack (top at right): (empty)

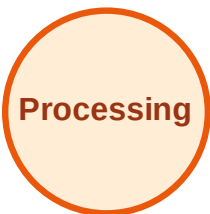
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

